- Town planning is not new idea or science in India.
- It is as old as its rich culture and tradition.
- Following are the development stages of town planning in India.

1. Ancient India- Vedic Period
2. Indus Valley civilization
3. Buddhist Period
4. Medieval Period (14th Century)
5. Moghul Period (1526 to 1707 A.D.)
6. Pre-independence or British Period (till 1947)
7. Post Independence Period.



TOPICS TO BE COVERED

- INDUS VALLEY CIVILIZATION
  - MOHENJADORO
  - HARAPPA
- EXTRACTS FROM CHANAKYA'S ARTHASASTRA
  - REGIONAL CONCEPTS
  - MANASARA VASTU SASTRA
    - 8 TOWN PLANNING CONCEPTS
- PLANNING CONCEPTS BEHIND
  - FATEHPUR SIKHRI
  - JAIPUR
  - CHANDIGARH
  - DELHI

Indus Valley Timeline

3300 BC	Small villages are set up in the area around the Indus Valley River.
2500 BC	People started to build planned cities at Mohenjo-Daro, Harappa and Lothal.
2400 - 1800 BC	The Indus Valley civilisation starts to grow with busy cities and trade.
1800 BC	Most of the Indus Valley cities fall into decline and later abandoned. We do not know why.
1856 AD	Railway engineers discover bricks that were made by the Indus Valley civilisation.
1921 AD	Harappa ruins were discovered.
1922 AD	Mohenjo-Daro's ruins are discovered.



What Is A Civilization?



Civilization Is The Progressive Phase Of All-round Development Of People Living In A Given Territory.



Growth Of Economy, Use Of Technology, Living In Cities.



Specialization In Skills, Script And Agriculture.

The Bronze Age Civilization

It Came After The Copper Age Civilization.

Bronze Is The Alloy Of Copper Bronze And Tin.

During This Period, Bronze Was Used.

Hence, This Time Period Got Its Name As Bronze Age Civilization.

Discovery Of The Civilization



Sir John Marshall Led His Team To Discover The Harappan Civilization.

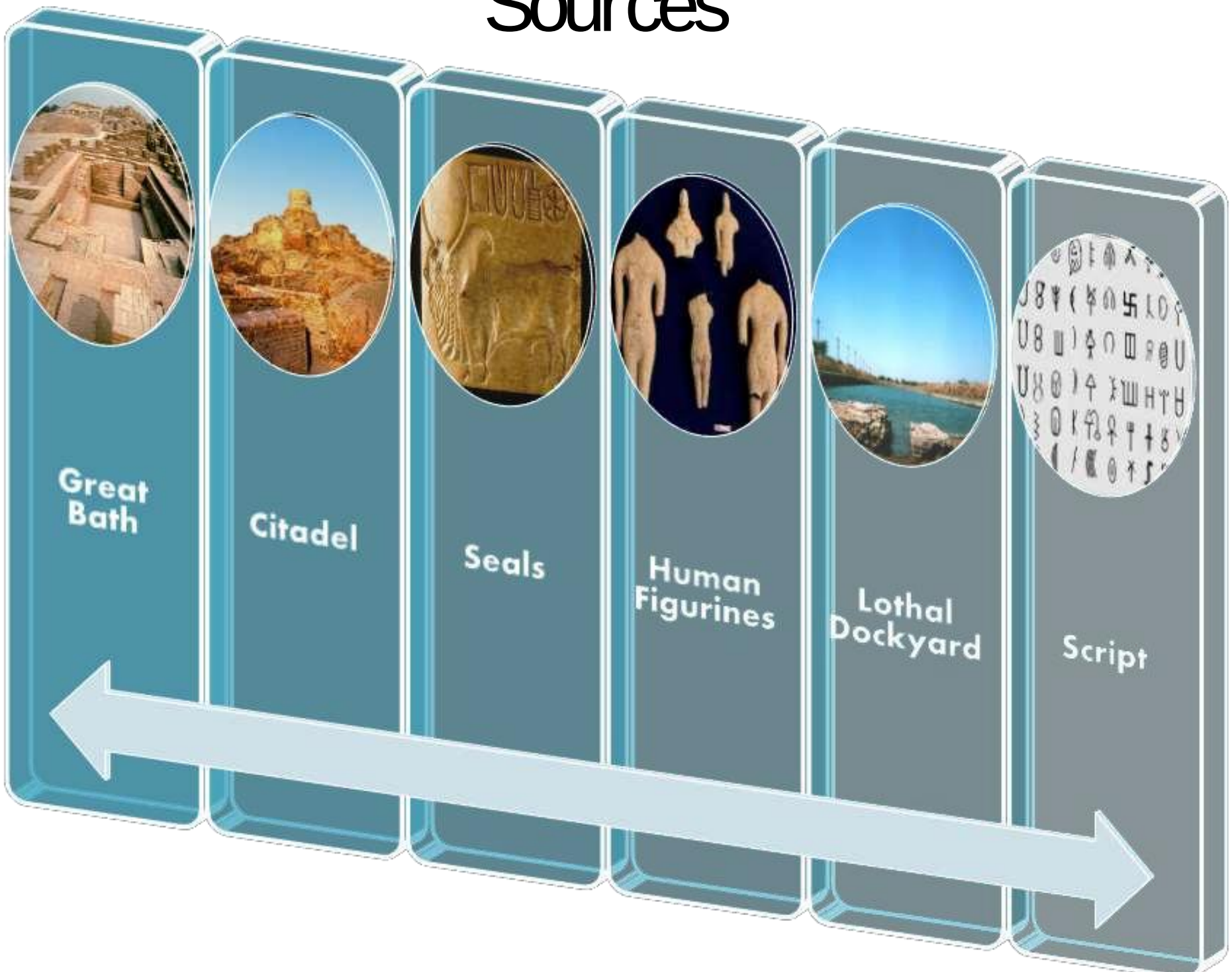


Dayaram Sahni Discovered The Ruins In Harappa In 1921.



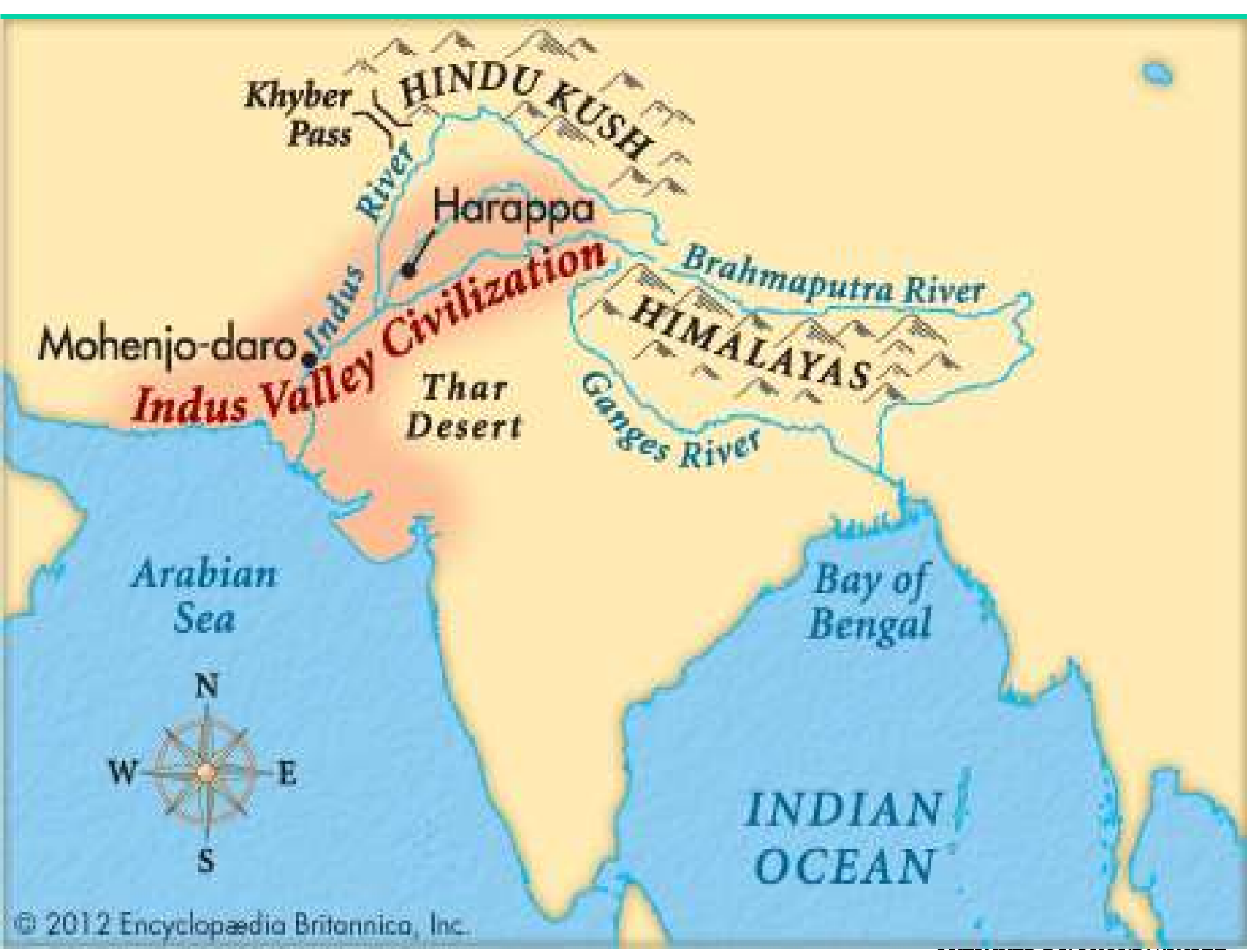
Rakhaldas Banerjee Discovered Mohenjo-Daro In 1922.

Sources

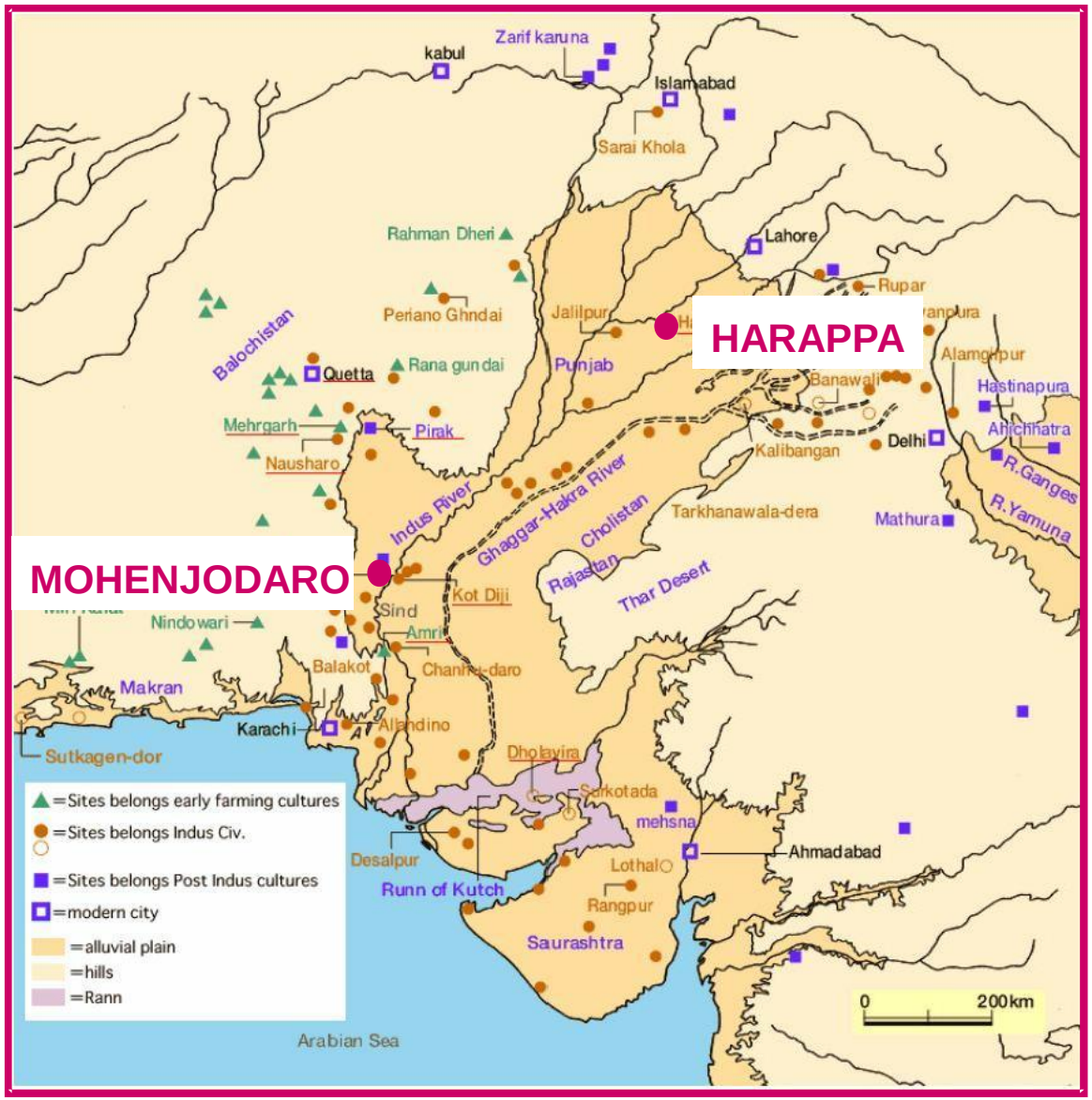

Great Bath
Citadel
Seals
Human Figurines
Lothal Dockyard
Script

Extent Of The Civilization

- Western Border - Suktagendor In South Baluchistan
- Eastern Border – Alamgirpur In Uttarpradesh
- Northern Border – Mandan In Jammu And Kashmir
- Southern Border – Bhagatrav In The Narmada Estuary


Khyber Pass
HINDU KUSH
Indus River
Harappa
Brahmaputra River
HIMALAYAS
Mohenjo-daro
Indus Valley Civilization
Thar Desert
Ganges River
Arabian Sea
Bay of Bengal
N
W
E
S
INDIAN OCEAN
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INDUS VALLEY CIVILIZATION



- Also referred to as **HARAPPAN CIVILIZATION & SARASWATHI SINDHU CIVILIZATION**
- Between **INDUS RIVER AND THE GHAGGAR-HAKRA RIVER** [Pakistan and North Western India]

INDUS VALLEY CIVILIZATION

TOWN PLANNING CONCEPT

- SOPHISTICATED & ADVANCED URBAN CULTURE
- STREETS IN PERFECT GRID PATTERNS IN BOTH MOHENJODORO & HARAPPA
- WORLD'S FIRST SANITATION SYSTEM
- INDIVIDUAL WELLS AND SEPARATE COVERED DRAINS ALONG THE STREETS FOR WASTE WATER
- HOUSES OPENED TO INNER COURTYARDS & SMALLER LANES
- IMPRESSIVE DOCKYARDS, GRANERIES, WAREHOUSES, BRICK PLATFORMS & PROTECTIVE WALLS
- MASSIVE CITADELS PROTECTED THE CITY FROM FLOODS & ATTACKERS
- CITY DWELLERS – TRADERS & ARTISANS
- ALL THE HOUSES HAD ACCESS TO WATER & DRAINAGE FACILITIES

INDUS VALLEY CIVILIZATION

DEVELOPMENT OF CITIES

- CITIES GREW OUT OF EARLIER VILLAGES THAT EXISTED IN THE SAME LOCALITY FOR < 100 YRS
- GREW IN SIZE & DENSITY AND SURROUNDED BY NUMEROUS TOWNS & VILLAGES
- CITIES INTERLINKED BY TRADE & ECONOMIC ACTIVITIES, RELEGIOUS BELIEFS, SOCIAL RELATIONS,ETC
- VAST AGRICULTURAL LANDS, RIVERS & FORESTS BY PASTORAL COMMUNITIES , FISHER FOLK AND HUNTERS SURROUNDED EACH CITY

INDUS VALLEY CIVILIZATION

DEVELOPMENT OF CITIES

- CLASSIFICATION OF TOWNS
  - Small villages / hamlets – 0 – 10 hectares
  - Large towns – 10- 50 hectares
  - Cities – 50 hectares
- IMPORTANT CITIES

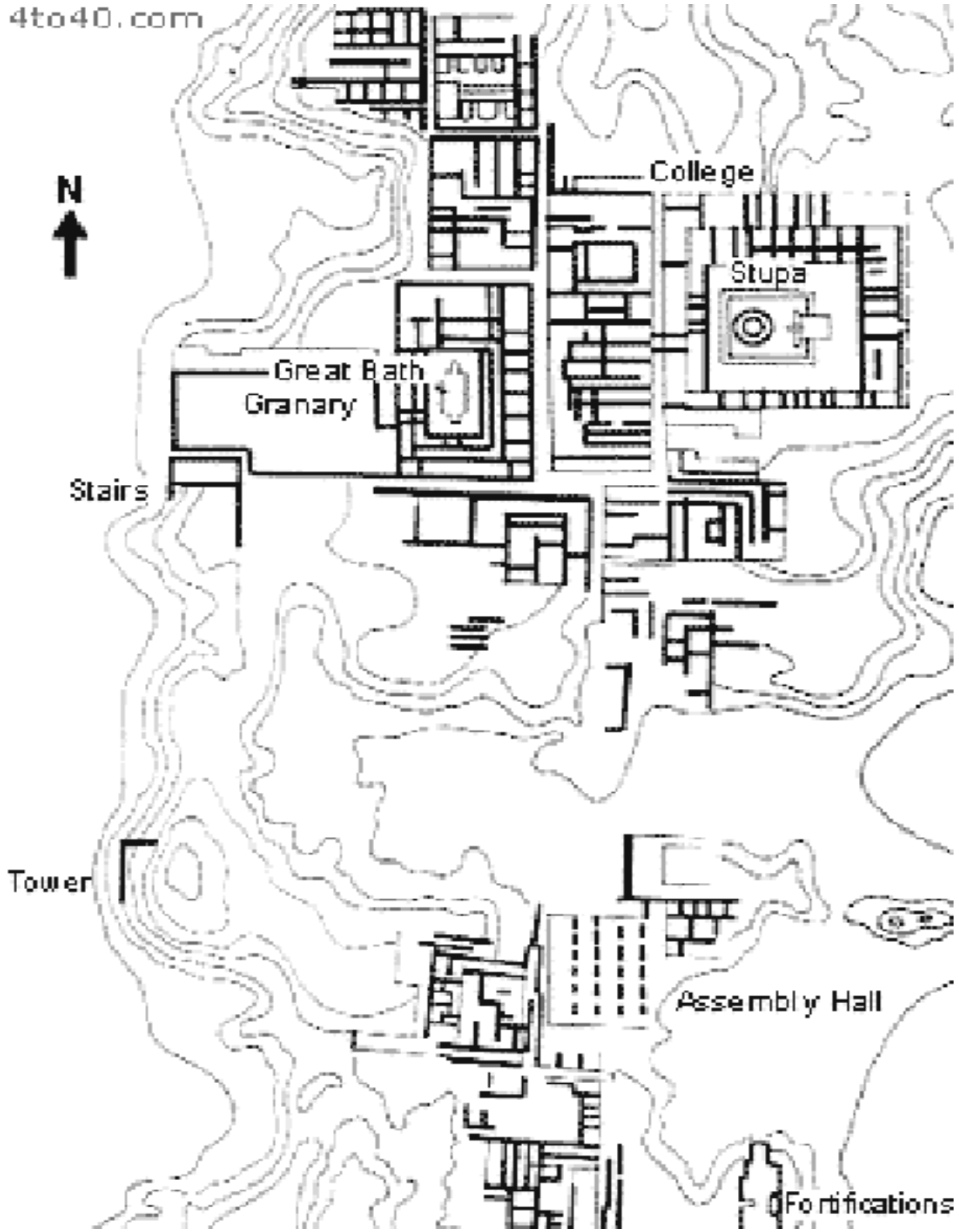
CITY	SIZE IN HECTARES	POPULATION
MOHENJODARO	200	35-41000
HARAPPA	150	23500
GANWERIWALA	80	
RAKHIGARHI	80	
DHOLAVIRA	100	
REHMAN DEHRI	22	12000



MOHENJODARO

- NO FORTIFICATION
- MAJOR STREETS IN NORTH SOUTH DIRECTION
- INTERSECTION AT RIGHT ANGLES
- STREETS WITHIN BUILT UP AREAS WERE NARROW
- DISTINCT ZONING FOR DIFFERENT GROUPS


SINDHU RIVER
UPPER CITY
LOWER CITY
1. Main Gate
2. Main Street
3. Cattle Bazaar
4. Market Square
5. Granary Square
6. Shrine Forest
7. Entry to Shrine
8. Arena
9. Upper City Gate
10. Maham's House
11. Great Bath
12. Senate
13. River Gate
14. North Gate
THE MAP OF
MOHENJO DARO
N



MOHENJODARO

SETTLEMENT DIVISIONS

- RELEGIOUS, INSTITUTIONAL & CULTURAL AREAS – AROUND MONASTERY & GREAT BATH IN THE WESTERN PART
- NORTH – AGRICULTURE & INDUSTRIES
- SOUTH – ADMINISTRATION, TRADE & COMMERCE

MOHENJODARO

CONSTRUCTION TECHNIQUES

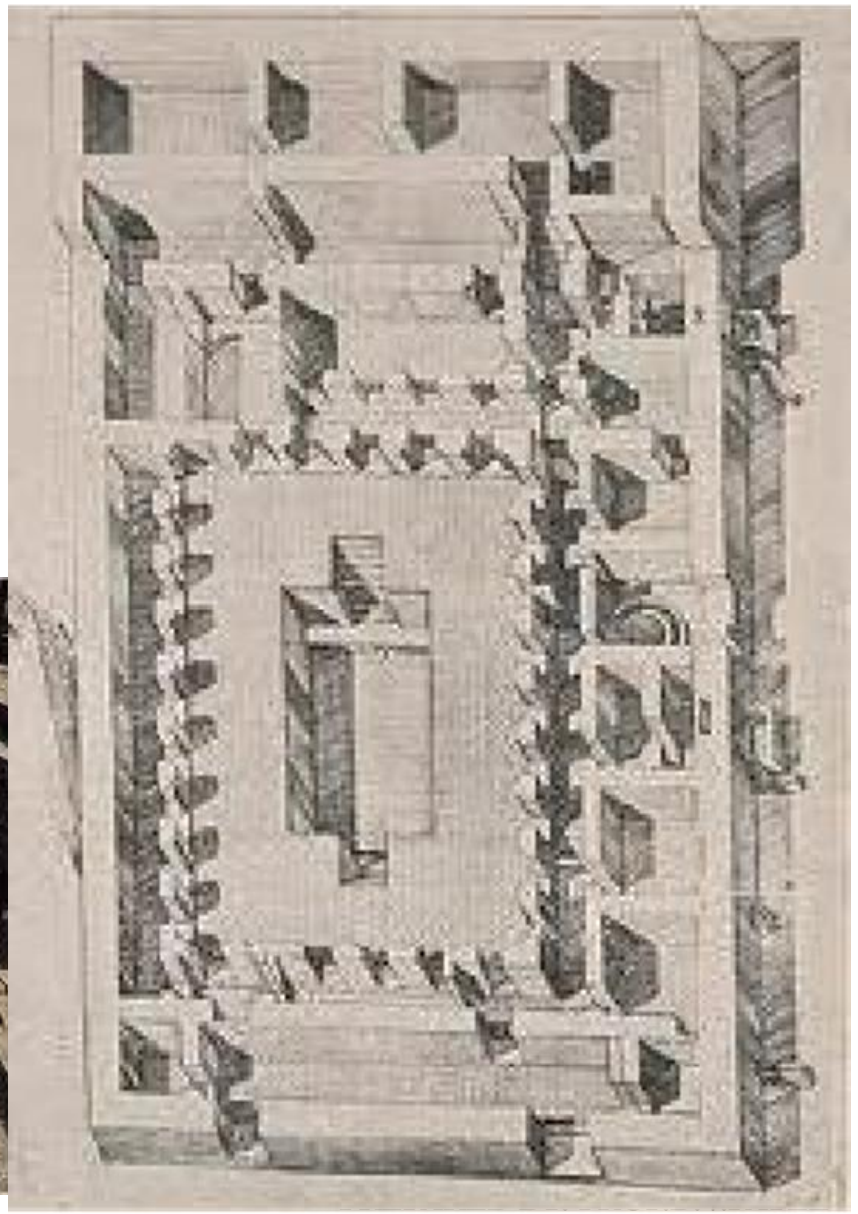
- BUILDINGS – MASONRY CONSTRUCTION BY SUN DRIED BRICKS
- RANGING FROM 2 ROOMS TO MANSIONS WITH MANY ROOMS
- UNDERGROUND SEWERAGE & DRAINAGE FROM HOUSES
- HELICAL PUMPS FOR PUMPING WATER IN GREAT BATH
- PRINCIPAL BUILDINGS – MONASTRY & BATH - INDICATING RELIGIOUS CULTURE

MOHENJODARO

GREAT BATH

- 12x7x3 M IN DIMENSIONS
- EARLIEST PUBLIC WATER TANK IN ANCIENT WORLD
- LEDGE EXTENDS FOR THE ENTIRE WIDTH OF POOL
- WATERTIGHT FLOOR – THICK LAYER OF BITUMEN
- FLOOR SLOPES IN SOUTH WEST CORNER WITH A SMALL OUTLET CONNECTING TO A BRICK DRAIN
- ROOMS LOCATED IN THE EAST



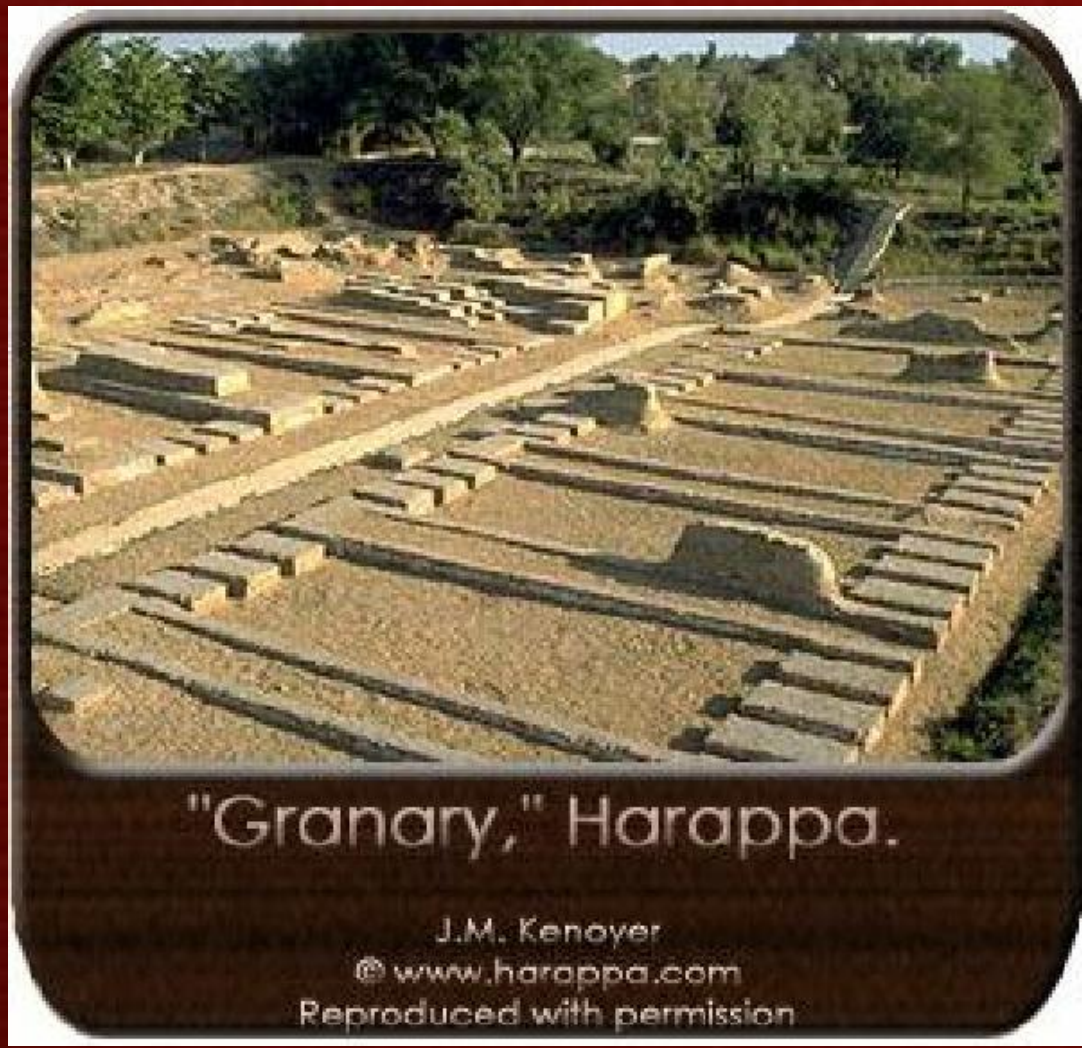


MOHENJODARO

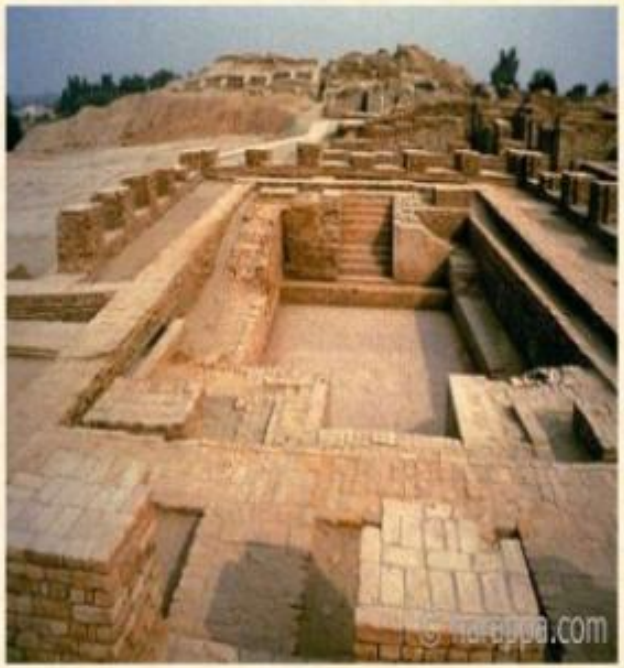
GRANARY

- 50x40 M IN DIMENSION, 4.5 M TALL
- MASSIVE MUD BRICK FOUNDATION
- 2 ROWS OF SIX ROOMS ALONG A CENTRAL PASSAGEWAY [7M WIDE & PAVED WITH BAKED BRICKS]
- EACH ROOM 15.2x6.1 M HAS 3 STEEPER WALLS WITH AIRSPACE BETWEEN
- A WOODEN SUPERSTRUCTURE SUPPORTED IN SOME PLACES BY LARGE COLUMNS WOULD HAVE BEEN BUILT ON TOP OF THE BRICK FOUNDATIONS, WITH STAIRS LEADING UP FROM THE CENTRAL PASSAGE AREA.
- SMALL TRIANGULAR OPENIGS – AIR DUCTS FOR FRESH AIR BENEATH HOLLOW FLOORS
- THE LARGE SIZE OF THE GRANARY PROBABLY INDICATES A HIGHLY DEVELOPED AGRICULTURAL CIVILIZATION

- In 1950, Sir Mortimer Wheeler identified one large building in Mohenjo-Daro as a "Great Granary".
- Certain wall-divisions in its massive wooden superstructure appeared to be grain storage-bays, complete with air-ducts to dry the grain.
- Granary is adjacent to the Great Bath.
- There were two wells which lie in an unsorted complex of walls which extend north-wards from the northern side of the Granary.



ANCIENT MOHEN-JODARO MODERN WORLD

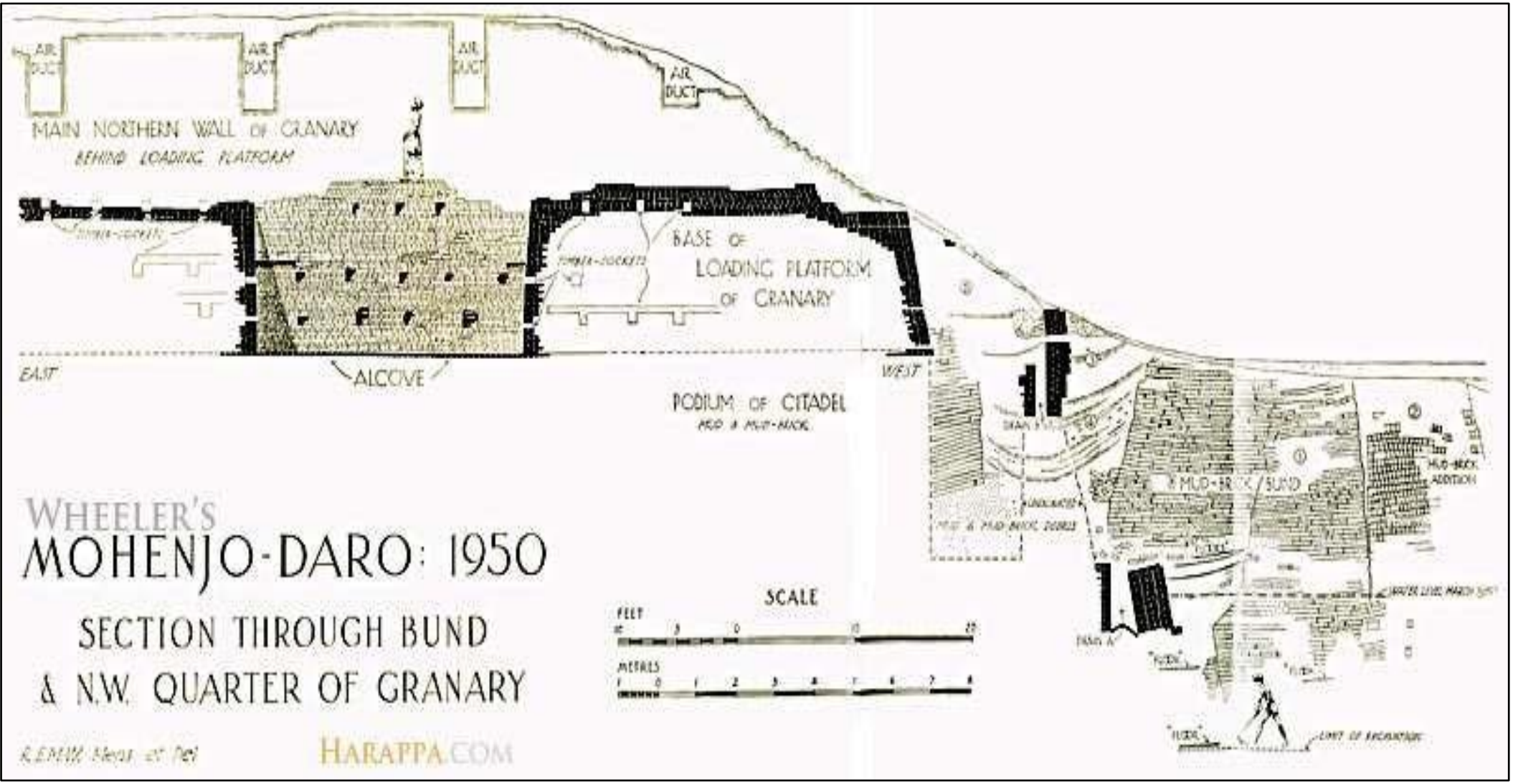




ANCIENT MOHEN-JODARO MORDERN WORLD







PREPARED BY: PROF.D.V.PATEL

HARAPPA



- 23000 POPULATION
- 150 HECTARES
- EARLIEST CITY MAY HAVE BEEN FORMED DURING THE KOT DIJI PHASE, I.E., 2800-2500 BC
- EARLIEST CITY COVERED AN AREA OF 25 HA.
- IT BECAME A CENTRE FOR TRADE NETWORKS EXTENDING FROM BALUCHISTAN AND AFGHANISTAN TO THE WEST OF THE SEACOAST IN THE SOUTH.
- TOWNS BUILT OVER RAISED MUD BRICK PLATFORMS



HARAPPA

TOWN PLANNING

- CITADEL MOUND AND LOWER TOWN SURROUNDED BY A **MASSIVE BRICK WALL.**
- CITADEL HAD **SQUARE TOWERS AND BASTIONS**.
- LARGE OPEN AREAS INSIDE THE GATEWAY MAY HAVE BEEN USED AS A **MARKET OR CHECKPOINT FOR TAXING GOODS COMING INTO THE CITY**
- OUTSIDE THE CITY WALLS A **CLUSTER OF HOUSES** MAY REPRESENT **TEMPORARY REST STOPS FOR TRAVELLERS AND CARAVANS**
- **NO DIVISION** OF THE SOCIETY IS REFLECTED IN THE LAYOUT OF THE CITY. SINCE LARGE PUBLIC BUILDINGS, MARKET AREAS, LARGE AND SMALL HOUSES AS WELL AS CRAFT WORKSHOPS HAVE BEEN FOUND IN THE SAME NEIGHBOURHOOD.
- **BARRACK-LIKE GROUP OF SINGLE-ROOMED TENEMENTS** WERE FOR THE POORER CLASSES

HARAPPA

TOWN PLANNING

- Basic house plans
  - single room tenements
  - houses with courtyards
- Houses - rooms on 3 sides opening into a central courtyard
- Nearly all large houses had private wells.
- Hearths (brick- or stone-lined fireplace or oven often used for cooking and/or heating) common in rooms.
- Bathrooms in every house with chutes leading to drainage channels.
- First floor bathrooms also built.
- Brick stairways provided access to the upper floors.
- Houses built with a perimeter wall and adjacent houses were separated by a narrow space of land.
- Granary with areas for threshing grains.
- Burnt bricks mainly used for drains, wells and bathrooms.
- Sun dried bricks used mainly for fillings.
- Timber used for flat roofs and as frames or lacing for brickwork

HARAPPA



ARCHITECTURE

- BUILDING MATERIALS - MUD BRICKS AND BAKED BRICKS, WOOD AND REEDS.
- THE AVERAGE SIZE OF THE BRICKS WAS 7 X 12 X 34 CM (FOR HOUSES) AND 10 X 20 X 40 CM FOR THE CITY WALLS. THE LARGER BRICKS HAVE A STANDARD RATIO OF 1:2:4.
- MUD BRICK, BAKED BRICK & WOOD OR STONE WERE USED FOR THE FOUNDATION AND WALLS OF THE HOUSES.
- THE DOORS ,WINDOWS WERE MADE FROM WOOD AND MAT.
- HOUSE FLOORS -HARD-PACKED EARTH
- BATHING AREAS AND DRAINS - BAKED BRICK AND STONE.
- ROOFS -WOODEN BEAMS COVERED WITH REEDS AND PACKED CLAY.
- LARGEST BUILDINGS MADE ENTIRELY OF WOOD.
- WINDOWS – SHUTTERS & LATTICE WORK

HARAPPA

ARCHITECTURE : LARGE PUBLIC STRUCTURES

- LARGE BUILDINGS -ADMINISTRATIVE OR RITUAL STRUCTURES.
- ACCESS ROUTES OR PROVIDED THOROUGHFARE FROM ONE AREA TO ANOTHER.
- MARKETS AND PUBLIC MEETINGS HELD IN LARGE OPEN COURTYARDS.
- HOUSES AND PUBLIC BUILDINGS GROUPED WITH SHARED WALLS AND FORMED LARGER BLOCKS & ACCESSED BY WIDE STREETS.
- MOST HOUSES HAD PRIVATE BATHS &TOILETS AS WELL AS PRIVATE WELLS.

HARAPPA

DRAINAGE SYSTEM

- WELLS AND RESERVOIRS - DRINKING AND BATHING.
- WELLS WERE LINED WITH SPECIALLY-MADE WEDGE-SHAPED BRICKS TO FORM A STRUCTURALLY SOUND CYLINDER.
- ROPES WERE USED TO LIFT THE WATER OUT, PROBABLY WITH LEATHER OR WOODEN BUCKETS.
- SOME NEIGHBOURHOODS HAD COMMUNAL WELLS.

HARAPPA

DRAINAGE SYSTEM

- BATHING PLATFORMS WITH WATER TIGHT FLOOR & DRAINS [OPEN OUT TO LARGER DRAINS IN STREETS] PROVIDED IN ROOMS ADJACENT TO THE WELLS.
- DRAINS AND WATER CHUTES IN THE UPPER STOREYS WERE OFTEN BUILT INSIDE THE WALL WITH AN EXIT OPENING JUST ABOVE THE STREET DRAINS.
- TAPERED TERRACOTTA DRAINPIPES WERE USED TO DIRECT WATER OUT TO THE STREET.
- MANY HOUSES HAD DISTINCT TOILETS, SEPARATE FROM THE BATH AREAS.
- COMMODES WERE LARGE JARS OR SUMP POTS SUNK INTO THE FLOORS AND MANY OF THEM CONTAINED A SMALL JAR.
- DRAINS COVERED WITH BAKED BRICKS OR DRESSED STONE BLOCKS. GARBAGE BINS WERE PROVIDED ALONG THE MAJOR STREETS.

Ancient India

Buddhist Period

PREPARED BY: PROF.D.V.PATEL

Buddhist Period (up to 320 A.D.)

- During the period of Emperor Chandragupta Maurya (321 B.C. – 185 B.C.), Kautilya a.k.a. Chanakya was the chief minister who wrote the famous '*Arthshastra*', a treatise (formal writing) on Town Planning.
- It statesthe regulation of zoning depending on communities, highways to be parallel to the main cardinal direction i.e. grid iron pattern.
- Width of Raj Marg to be not less than 30ft.
- It clearly shows that the art of Town Planning had made much progress as long back as third century B.C.

- The excavations carried out at Patliputra (modern Patna) the capital of Magadha, show evidence of advance knowledge of town planning.
- The capital was laid on grid-iron pattern consisting of 16 sectors.
- Most of the houses had gardens with wells and ponds. Waste water was carried out by means of underground drainage.
- The city as long as 16 km and 3.5 km wide was surrounded by a deep moat 180m wide and further protected by ramparts stretching more than 40 km long studded with 64 gates and 570 towers.

REGIONAL CONSIDERATION

CHANAKYA'S ARTHASASTHRA

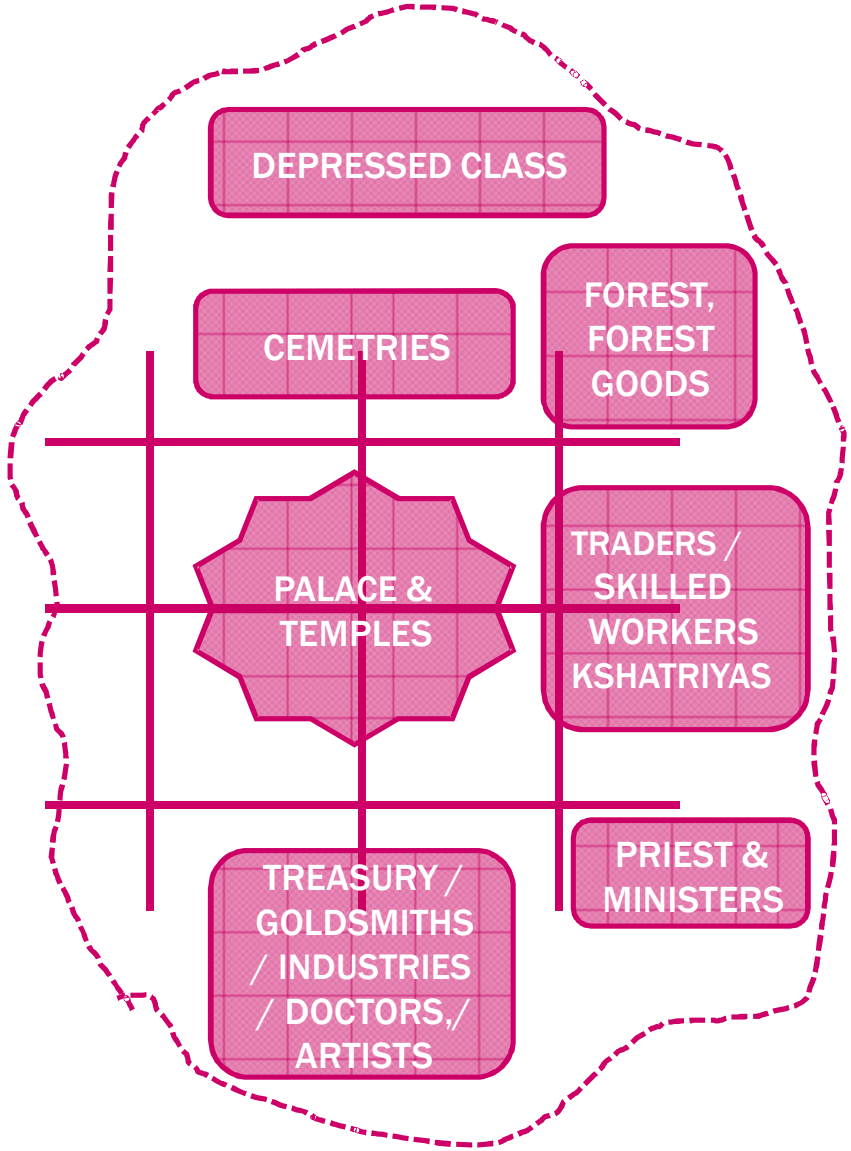
- CONGESTED TOWN, SHOULD BE FREED OF SURPLUS POPULATION, WHICH SHOULD THEN BE HOUSED IN A NEW PLACE.
- TOWNS POSITIONED TO HELP EACH OTHER.
- 'SANGRAHAN' (COLLECTION REGISTER / TAX COLLECTOR) - 10 VILLAGES, 'SARVATIK' AMONG 200, 'DRONAMUKH' (CHIEF) AMONG 400 AND 'STHANIYA' AMONG 800
- MIGRATED PEOPLE IN NEW SETTLEMENT EXEMPTED FROM PAYMENT OF TAXES FOR SOME YEARS.
- NEW VILLAGE –
  - HIGHER PROPORTION OF AGRICULTURISTS AND SHUDRAS.
  - MARKET - SALE OF GOODS RECEIVED FROM TRADERS ON HIGHWAYS.
  - DAMS - CONSTRUCTED OVER RIVERS NALAS.
  - TEMPLES AND GARDENS SHOULD BE PROVIDED.
  - ARRANGEMENTS FOR THE AGED, THE CHILDREN AND INFORMAL PERSONS.
  - CEREALS AND WEALTH WILL GROW IF THE AGRICULTURISTS ARE KEPT BUSY. ATTEMPTS SHOULD BE MADE TO PROTECT AND INCREASE QUARRIES, FORESTS AND CANALS.

PREPARED BY: PROF.D.V.PATEL

TOWN PLANNING

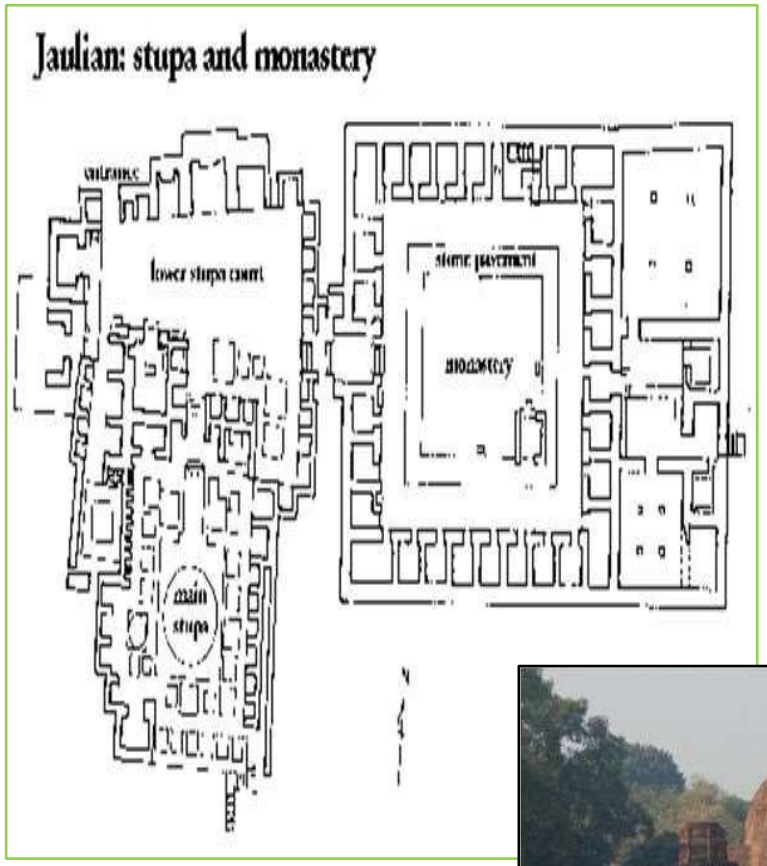
- A CITY - LOCATED CENTRALLY TO FACILITATE TRADE AND COMMERCE.
- THE SITE - LARGE IN AREA, AND NEAR A PERENNIAL WATER BODY .
- SHAPE - CIRCULAR, RECTANGULAR OR SQUARE AS WOULD SUIT THE TOPOGRAPHY.
- SEPARATE AREAS FOR MARKETING DIFFERENT GOODS.
- WALL AROUND THE TOWN, - 6 DANDAS HIGH AND 12 DANDAS WIDE. BEYOND THIS WALL THERE SHOULD BE THREE MOATS OF 14', 12'AND 10' WIDE TO BE CONSTRUCTED FOUR ARM-LENGTHS APART. DEPTH – 3/4TH OF WIDTH.
- THREE-EAST WEST AND THREE NORTH – SOUTH ROADS, SHOULD DIVIDE THE TOWN.
- THE MAIN ROADS SHOULD BE 8 DANDAS WIDE AND OTHER ROADS 4 DANDAS WIDE.
- 1 WELL FOR 10 HOUSES.

CHANAKYA'S ARTHASASTHRA



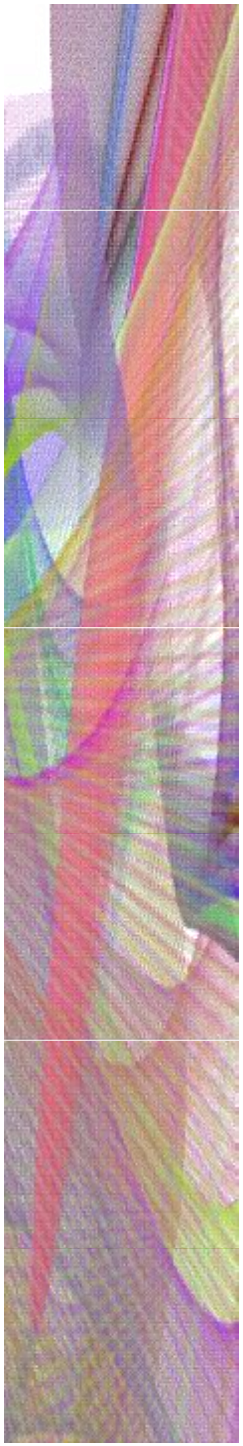
PREPARED BY: PROF.D.V.PATEL

Ancient India Buddhist Period









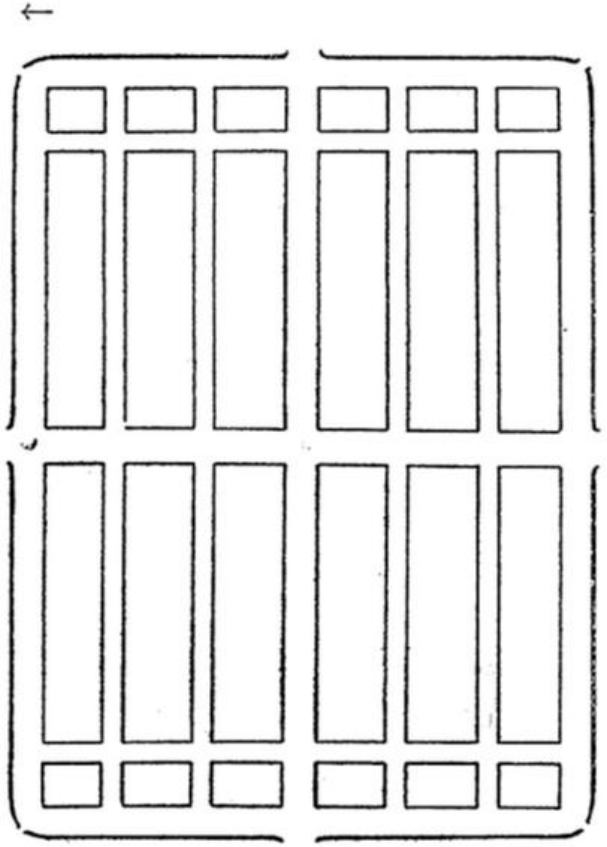
ANCIENT TOWN CLASSIFICATION

- Dandaka
- Sarvathobhadra
- Nandyavarta
- Padmaka
- Swastika
- Prastara
- Karmuka
- Chaturmukha

ANCIENT TOWN CLASSIFICATION

Dandaka

- STREETS ARE STRAIGHT AND CROSS EACH OTHER AT RIGHT ANGLES AT THE CENTRE
- VILLAGE HAS 4 GATES ON FOUR SIDES
- VILLAGE IS RECTANGULAR / SQUARE
- WIDTH OF THE STREET VARIES FROM ONE - FIVE DANDA
- 2 TRANSVERSE STREET AT THE EXTREMITIES HAVE SINGLE ROW OF HOUSES
- THE VILLAGE OFFICES LOCATED IN THE EAST.
- THE FEMALE DEITY/ CHAMADEVATA - LOCATED OUTSIDE THE VILLAGE AND THE MALE DEITIES IN THE NORTHERN PORTION

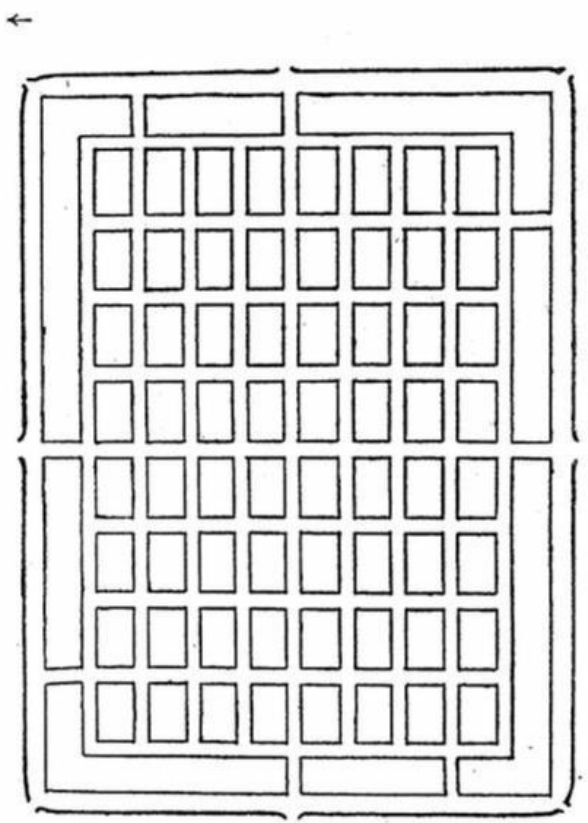


DANDAKA

ANCIENT TOWN CLASSIFICATION

SARVATOBHADRA

- This type of town plan is applicable to larger villages and towns, which have to be constructed on a square sites.
- According to this plan, the whole town should be fully occupied by houses of various descriptions and inhabited by all classes of people.
- The temple dominates the village

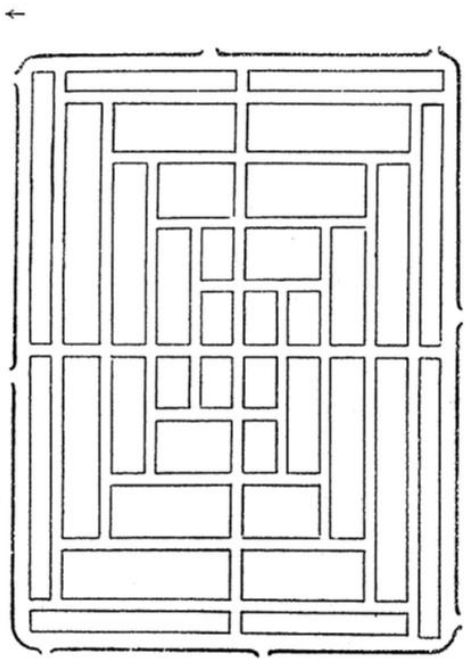


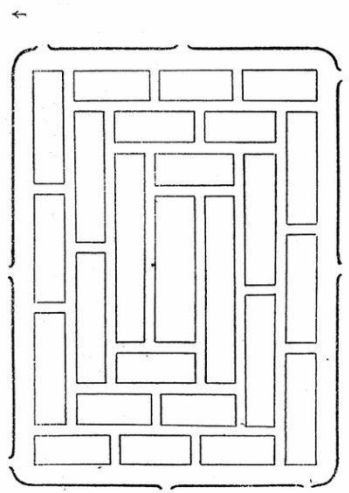
SARVATOBHADRA

ANCIENT TOWN CLASSIFICATION

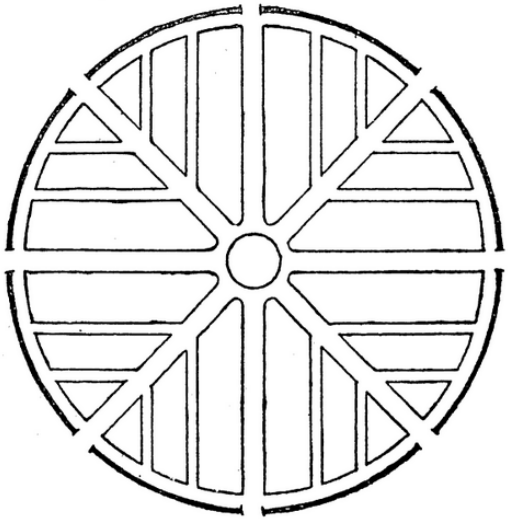
NANDYAVARTA

- This plan is commonly used for the construction of towns and not for villages.
- It is generally adopted for the sites either circular or square in shape, 3000 – 4000 HOUSES
- The streets run parallel to the central adjoining streets with the temple of the presiding deity in the center of the town.
- “Nandyavarta” is the name of a flower, the form of which is followed in this layout.





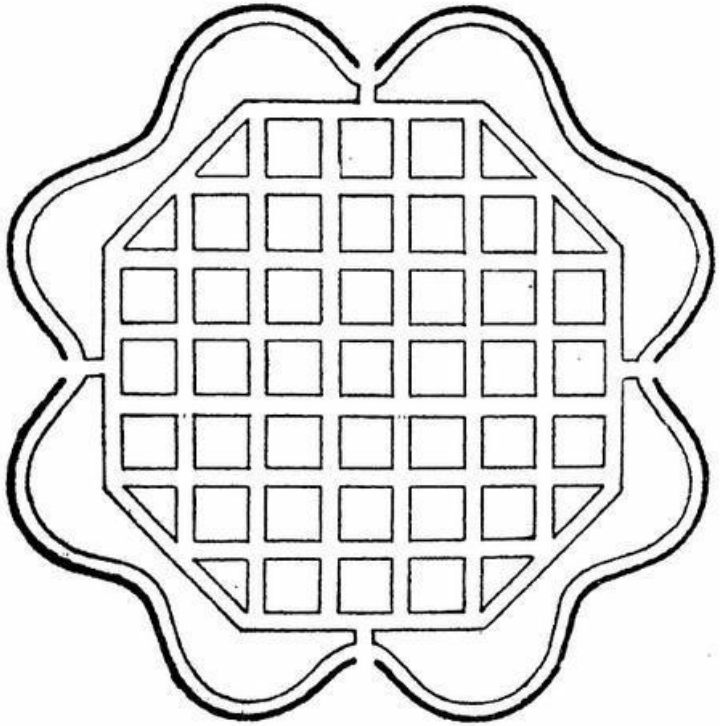
NANDYĀVARTA
(in *Sthaṇḍila*)



ANCIENT TOWN CLASSIFICATION

PADMAKA

- This type of plan was practiced for building of the towns with fortress all round.
- The pattern of the plan resembles the petals of lotus radiating outwards from the center.
- The city used to be practically an island surrounded by water, having no scope for expansion

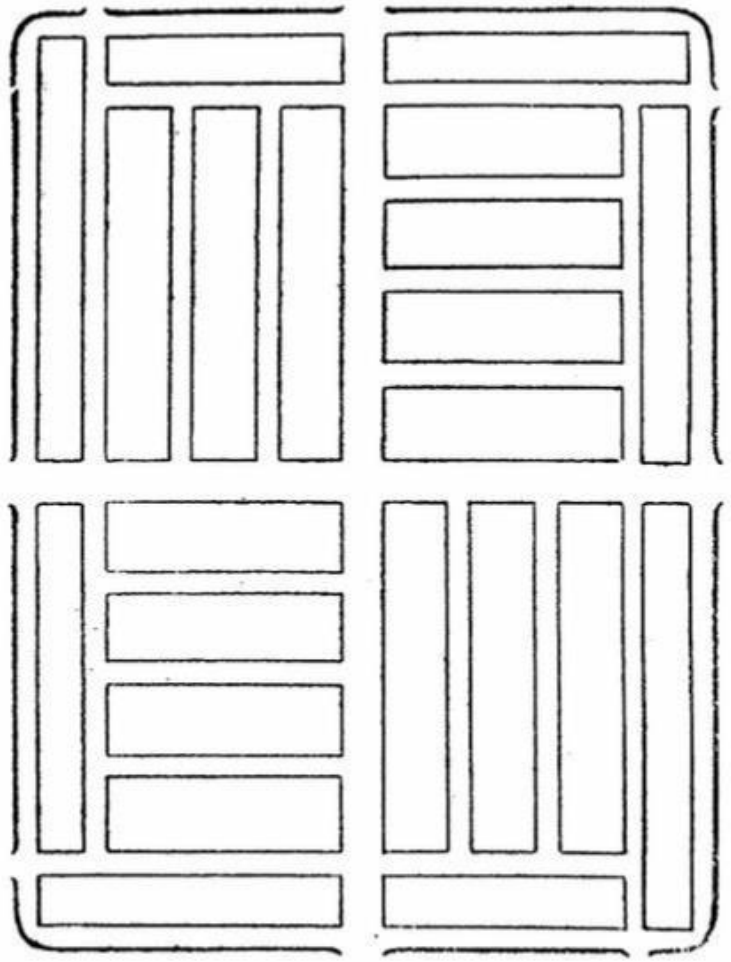


PADMAKA

ANCIENT TOWN CLASSIFICATION

SWASTIKA

- Swastika type of plan contemplates some diagonal streets dividing the site into certain rectangular plots.
- The site need not be marked out into a square or rectangle and it may be of any shape.
- A rampart wall surrounds the town, with a moat at its foot filled with water.
- Two main streets cross each other at the center, running south to north and west to east.

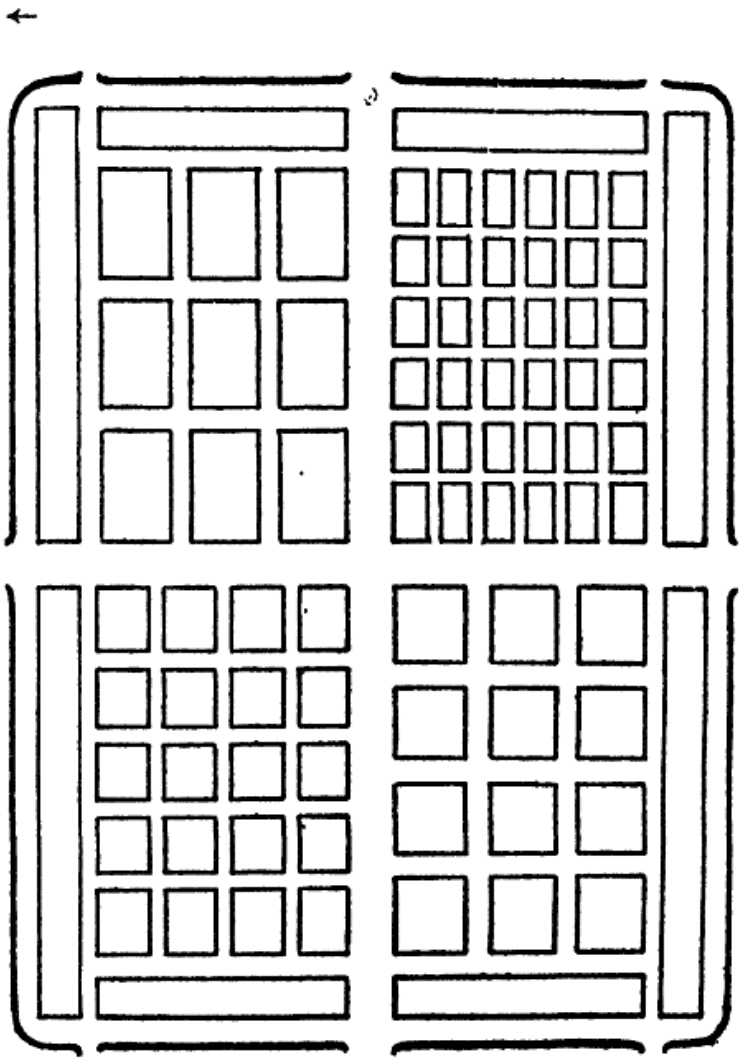


SWASTIKA

ANCIENT TOWN CLASSIFICATION

PRASTARA

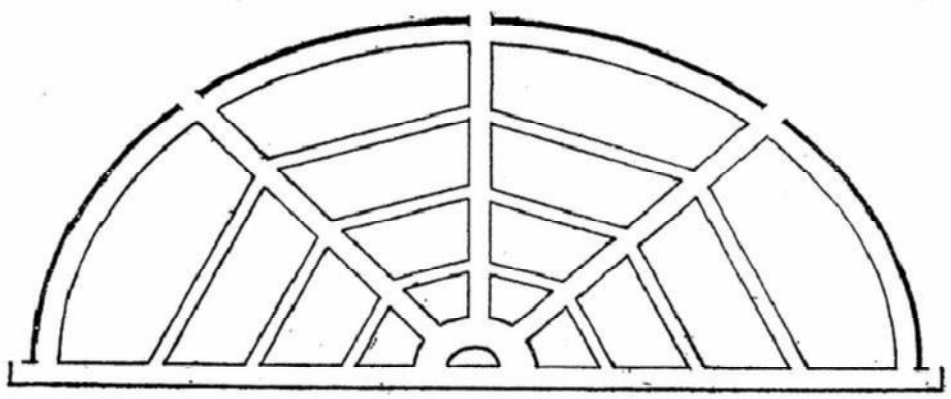
- The characteristic feature of this plan is that the site may be either square or rectangular but not triangular or circular.
- The sites are set apart for the poor, the middle class, the rich and the very rich, the sizes of the sites increasing according to the capacity of each to purchase or build upon.
- The main roads are much wider compared to those of other patterns.
- The town may or may not be surrounded by a fort.



ANCIENT TOWN CLASSIFICATION

KARMUKA

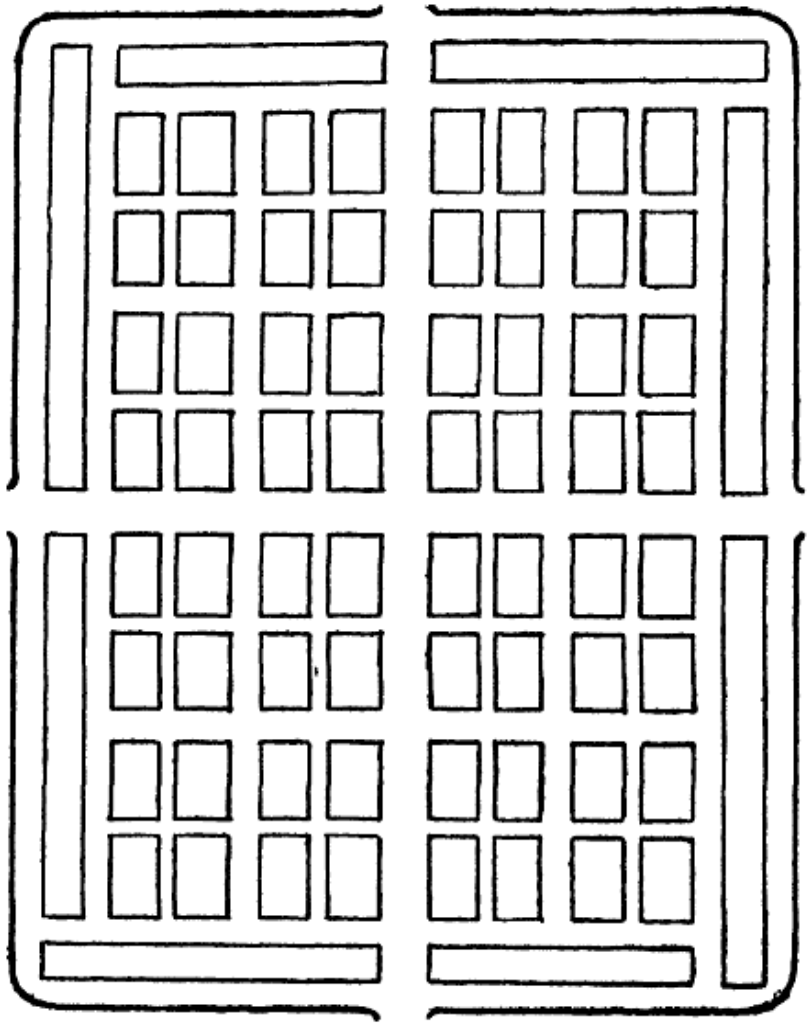
- This plan is suitable for the place where the site of the town is in the form of a bow or semi-circular or parabolic and mostly applied for towns located on the seashore or riverbanks.
- The main streets of the town run from north to south or east to west and the cross streets run at right-angles to them, dividing the whole area into blocks.
- The presiding deity, commonly a female deity, is installed in the temple build i



ANCIENT TOWN CLASSIFICATION

CHATURMUKHA

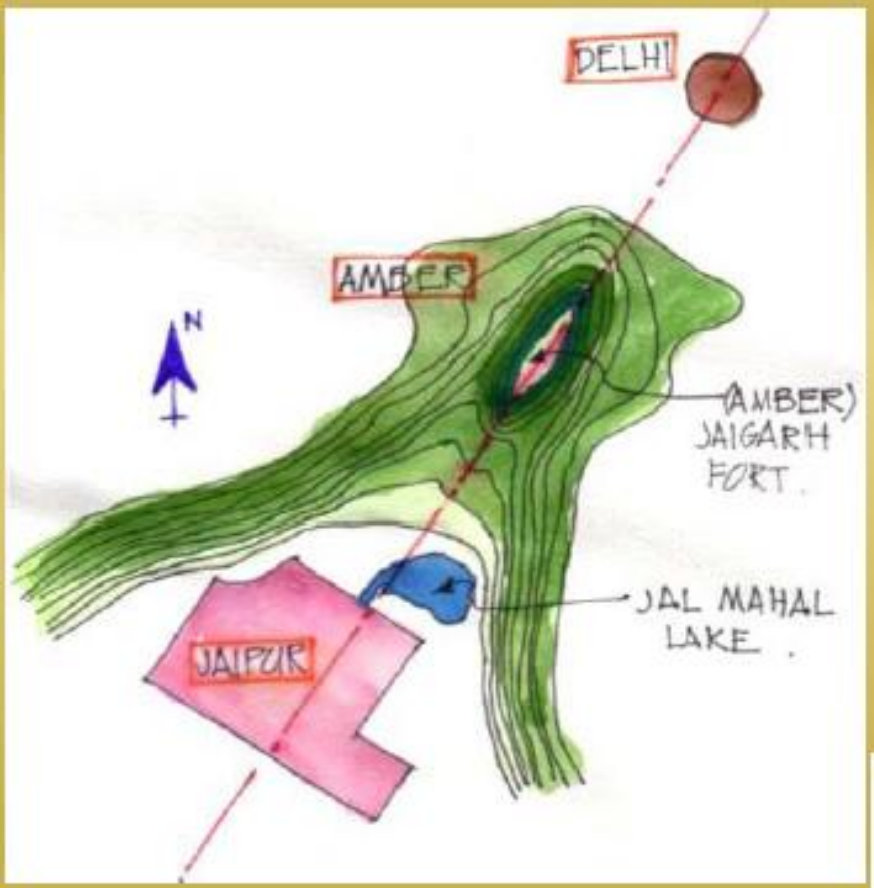
- Chaturmukha type of plan is applicable to all towns starting from the largest town to the smallest village.
- The site may be either square or rectangular having four faces.
- The town is laid out east to west lengthwise, with four main streets.
- The temple of the presiding deity will be always at the center



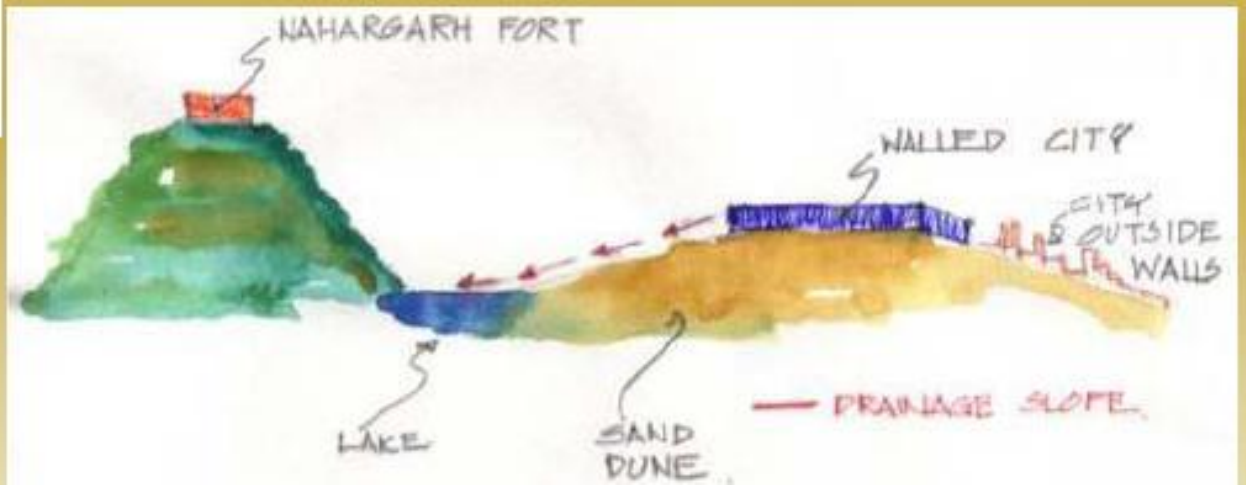
Medieval Period (14th Century)

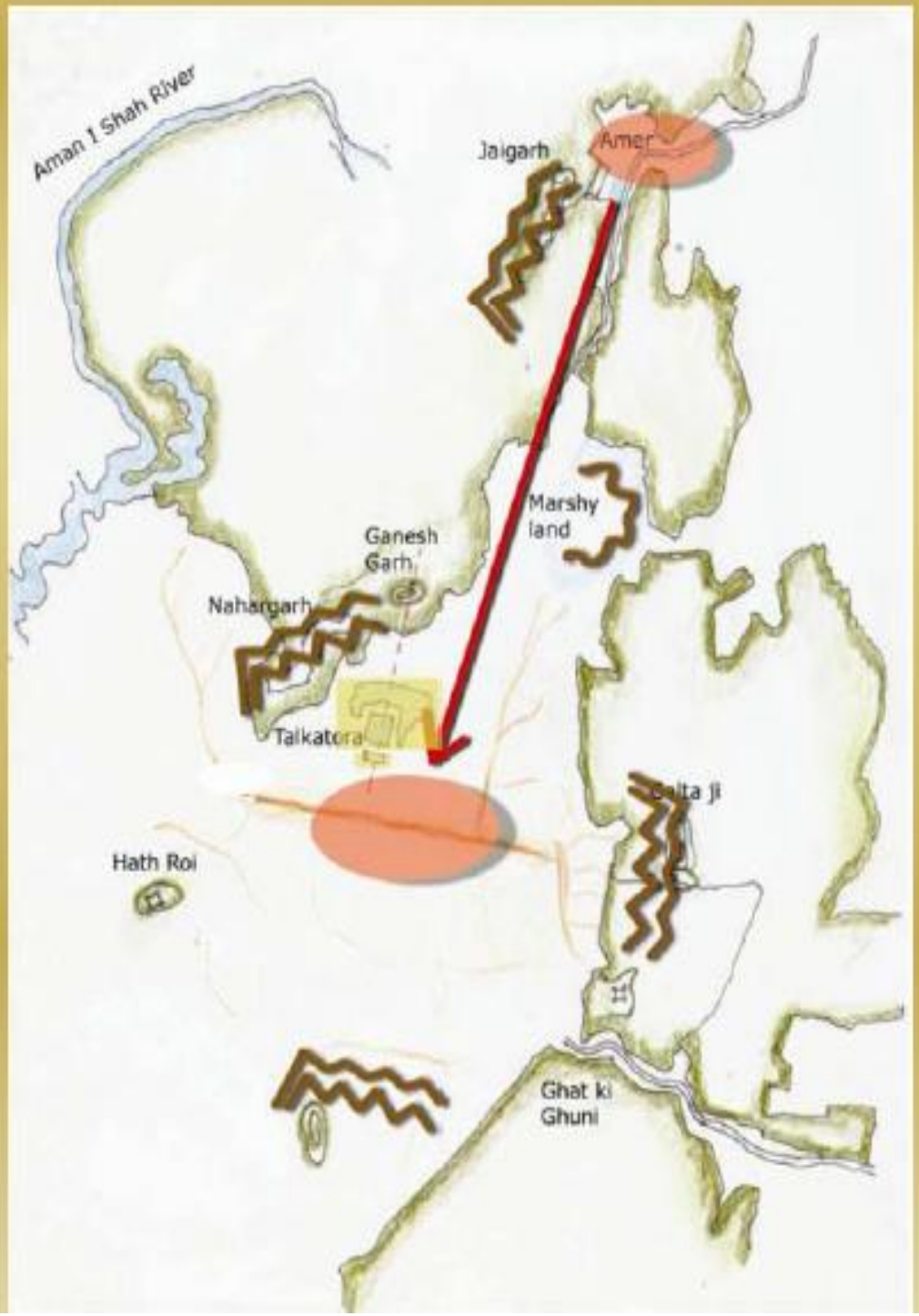
PREPARED BY: PROF.D.V.PATEL

Reasons for Maharaja Sawai Jai Singh to change his capital from Amber to Jaipur



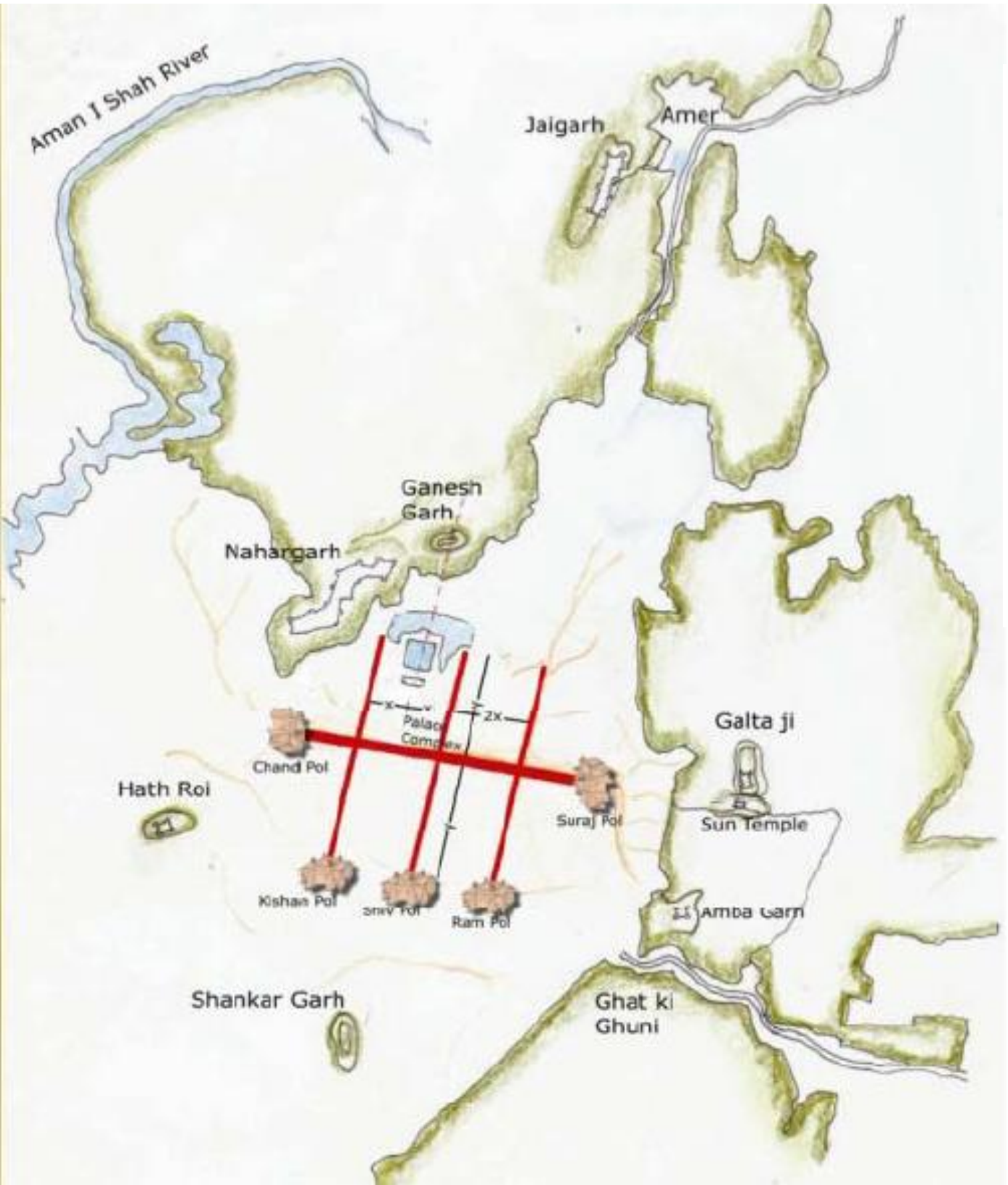
- Military Reasons
- Geographical Reasons





- The need of a new capital for 18th century Dhoondhar as the earlier one of Amber built on a hill was getting congested.

- Sawai Raja Jai Singh's vision of the new capital as a strong political statement at par with Mughal cities and as a thriving trade and commerce hub for the region.







PREPARED BY: PROF.D.V.PATEL

WHY PINK?



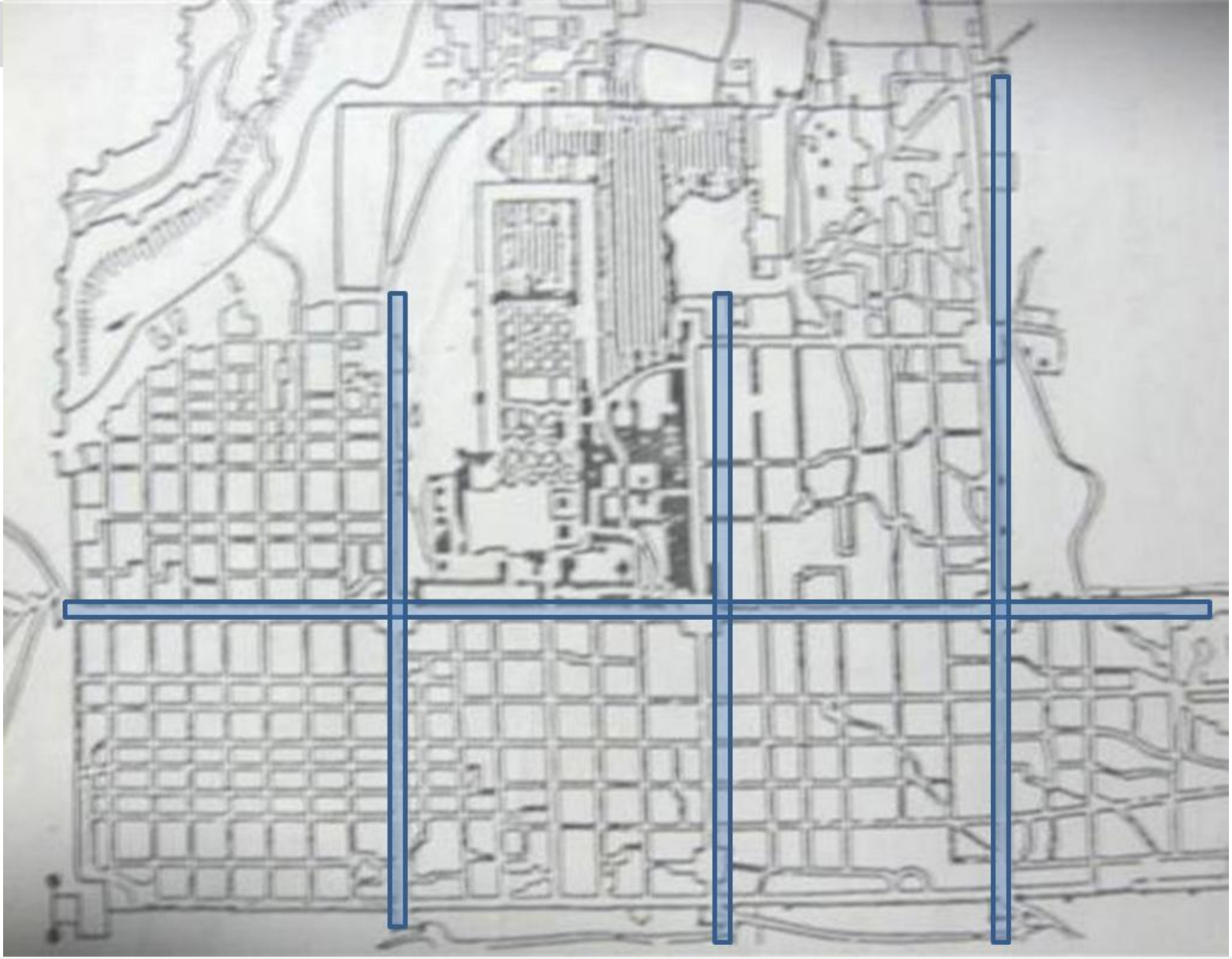


Using pink to cut down glare

Jai Singh II's apparent devotion to Lord Shiva (whose favourite colour is reputedly terra cotta)

Pink is the traditional color of hospitality, and the city was freshly painted and paved with pink gravel to warmly welcome Edward VII for his visit here in 1876.

ROAD NETWORK



Jaipur's road network follows a definite hierarchy. The major east-west and north-south road ,form the sector boundaries and are called Rajmarg as they lead to the city gates. These measure 33m. wide.

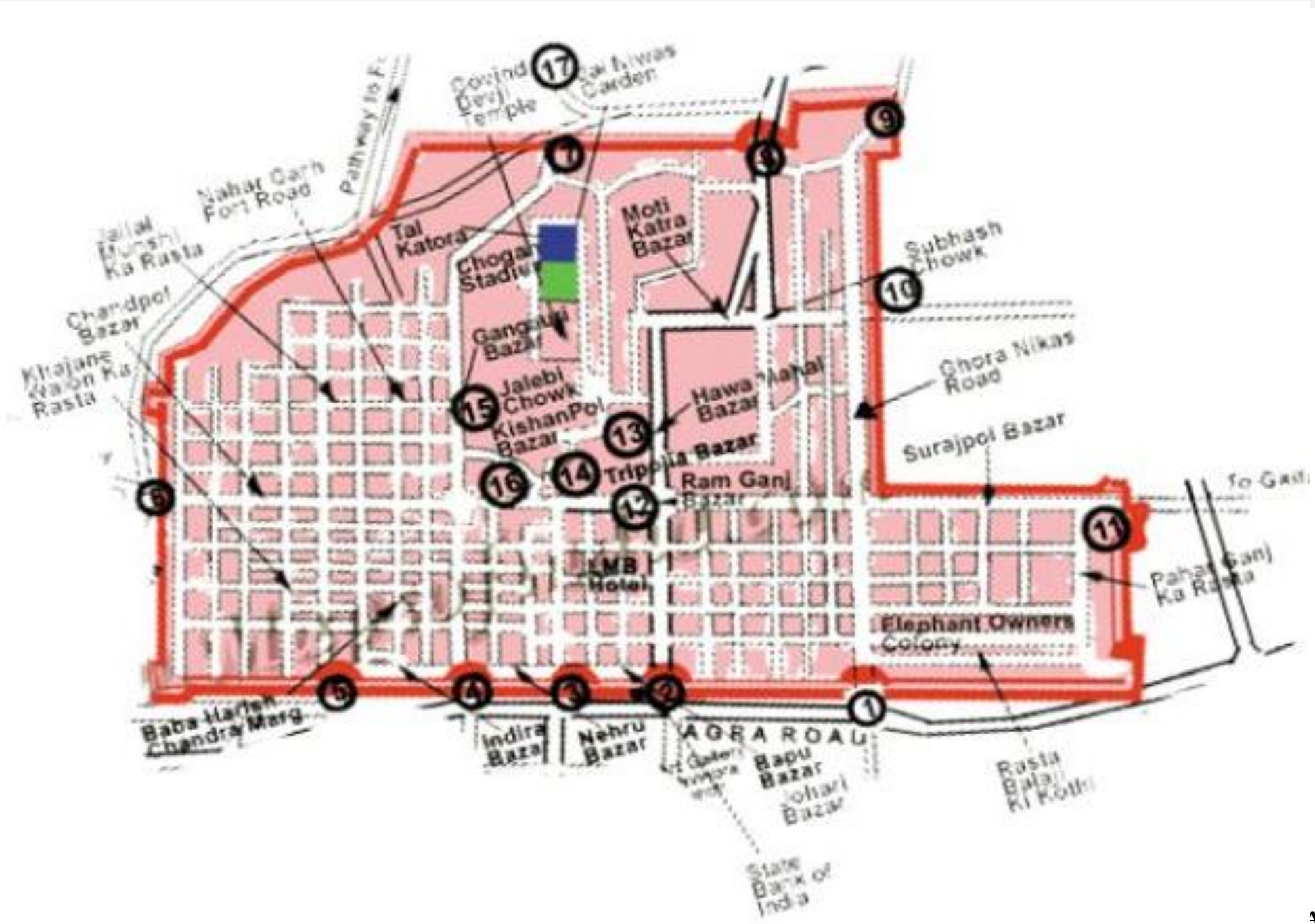
Next there is a network of 16.5m wide which runs north-south in each sector linking the internal areas of the sectors to the major activity spine.

PUBLIC SPACES



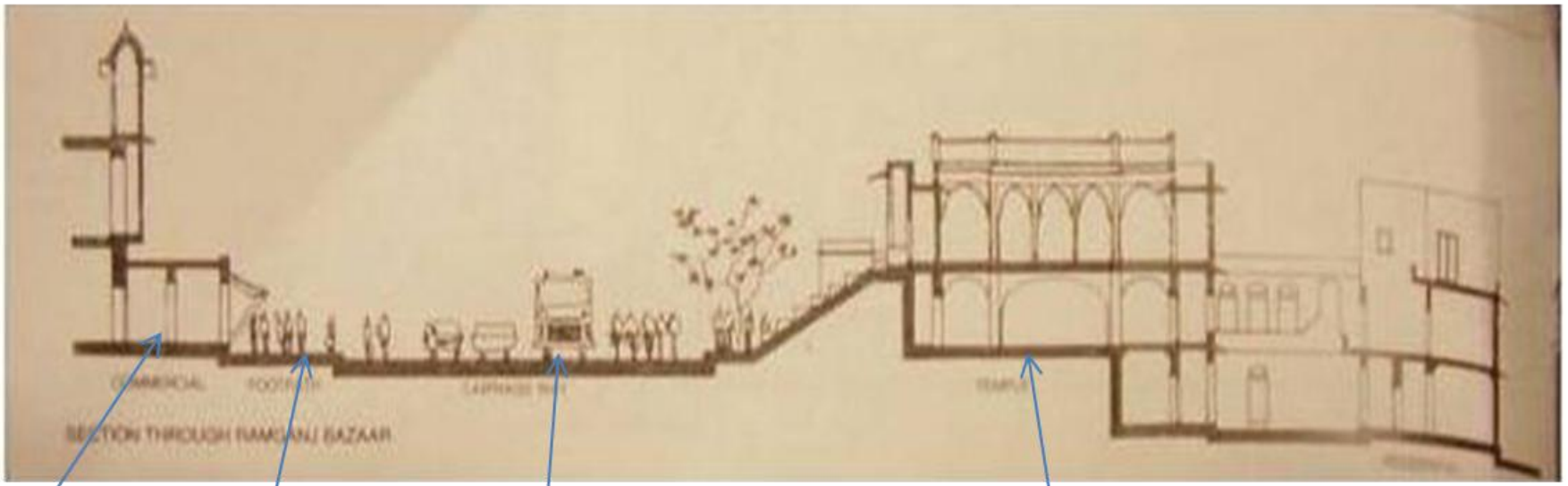
CHAUPAR – It's a square that occurs at the intersection of east west roads with three north south roads. Each chaupar is around 100m x 100m. Were used for public gathering on festive occasions. The distance between two chaupars is about 700m which is ideal for pedestrian movement. It has controlled façade treatment enveloping it.

BAZAARS - Originally only four bazaars were planned for the city. These were later named as Johri bazaar, Sireh Deori Bazaar, Kishan pole Bazaar & Gangori Bazaar



ATEL

Section through Ram Ganj Bazaar



Commercial

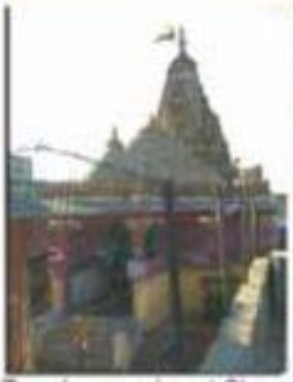
Footpath

Road

Temple

TYPOLOGY OF TEMPLES

Temples with Shikhara

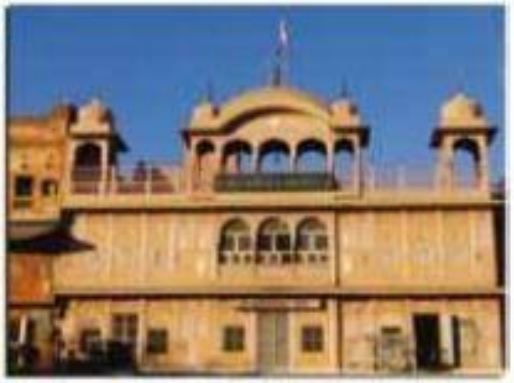




Temples opening at Chaupar

Temples without Shikhara – Haveli type temples





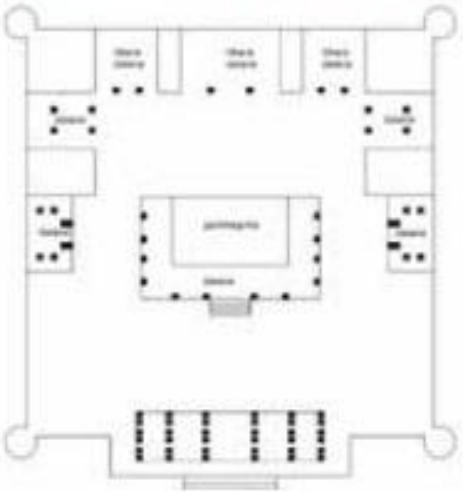
Approached by a flight of steps from the street

Temples with Baradari

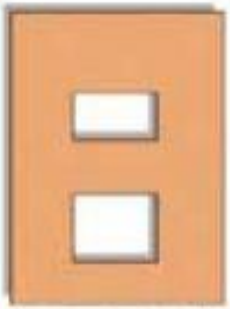




Govind Dev Temple



Temples with double courtyard





Shri Digamber Jain Mandir, Sanghji, Manharon Ka Rasta, Chowkri Modi Khana

Temples with single courtyard





Shri Digamber Jain Mandir, Sangaken Ka Rasta, Chowkri Modi Khana

Temples with no courtyard





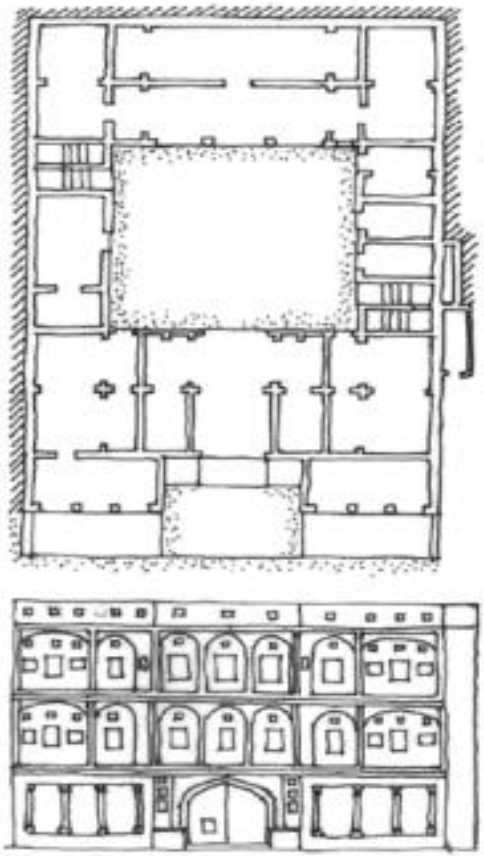
Imleshwar Vir Hanuman Mandir, Sankri Gali, Chowkri Modi Khana

Each mohalla (cluster of houses) has its own temple presided over by the deity most appropriate for their prosperity and protection. A relationship between temples and wells (both constituting ritual spaces) can be observed inside the chowkries in the layout of the sectors.

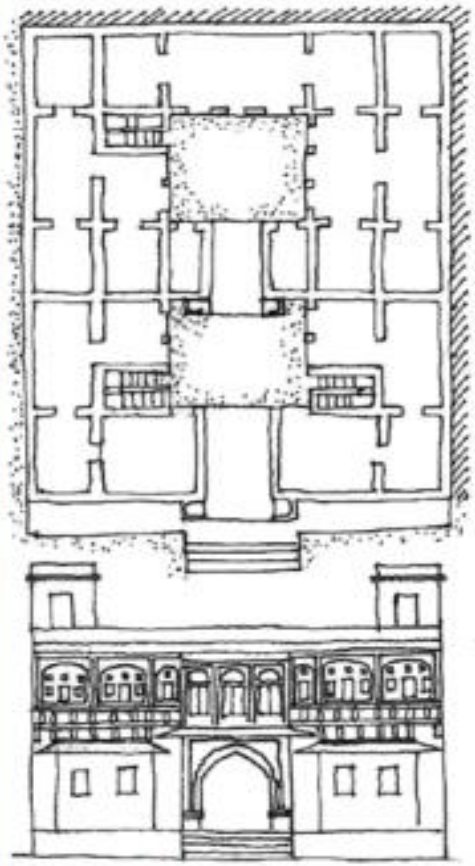
TYPOLOGY OF HAVELIS

The havelis of Jaipur range from a single courtyard house form to an assemblage of multiple courts, depending on the status of the owner and number of family members.

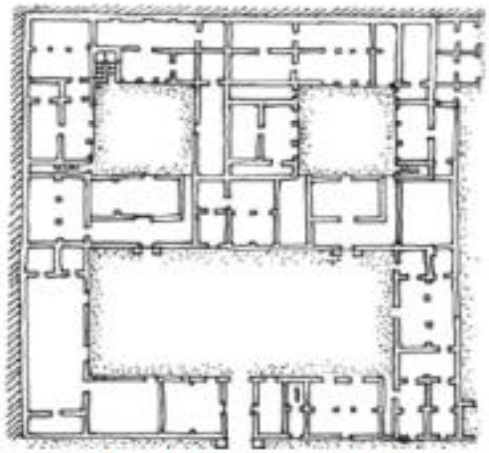
Majority of the havelis have one or two courtyards.



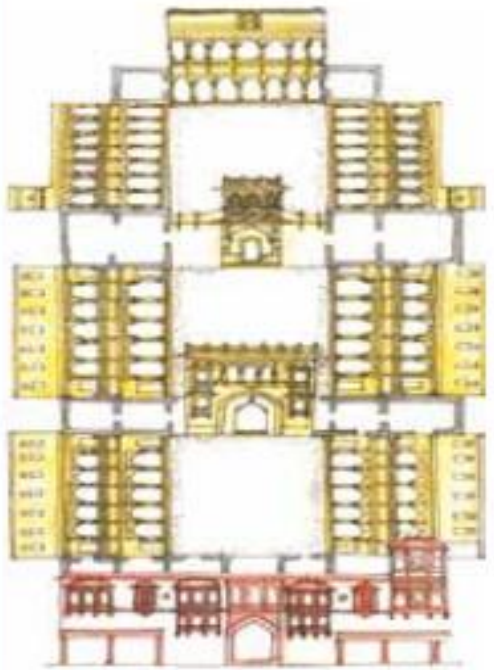
The Tatterkhana House, Jaipur, reflects a single family cluster consisting of a single family and its zenana (wing occupied by ladies of the house) and infrastructure of servants



The Shivdin Haveli, Jaipur with typical two court planning



Bhatt Haveli with three courtyards



Natani Haveli with multiple courts, Jaipur

Moghul Period

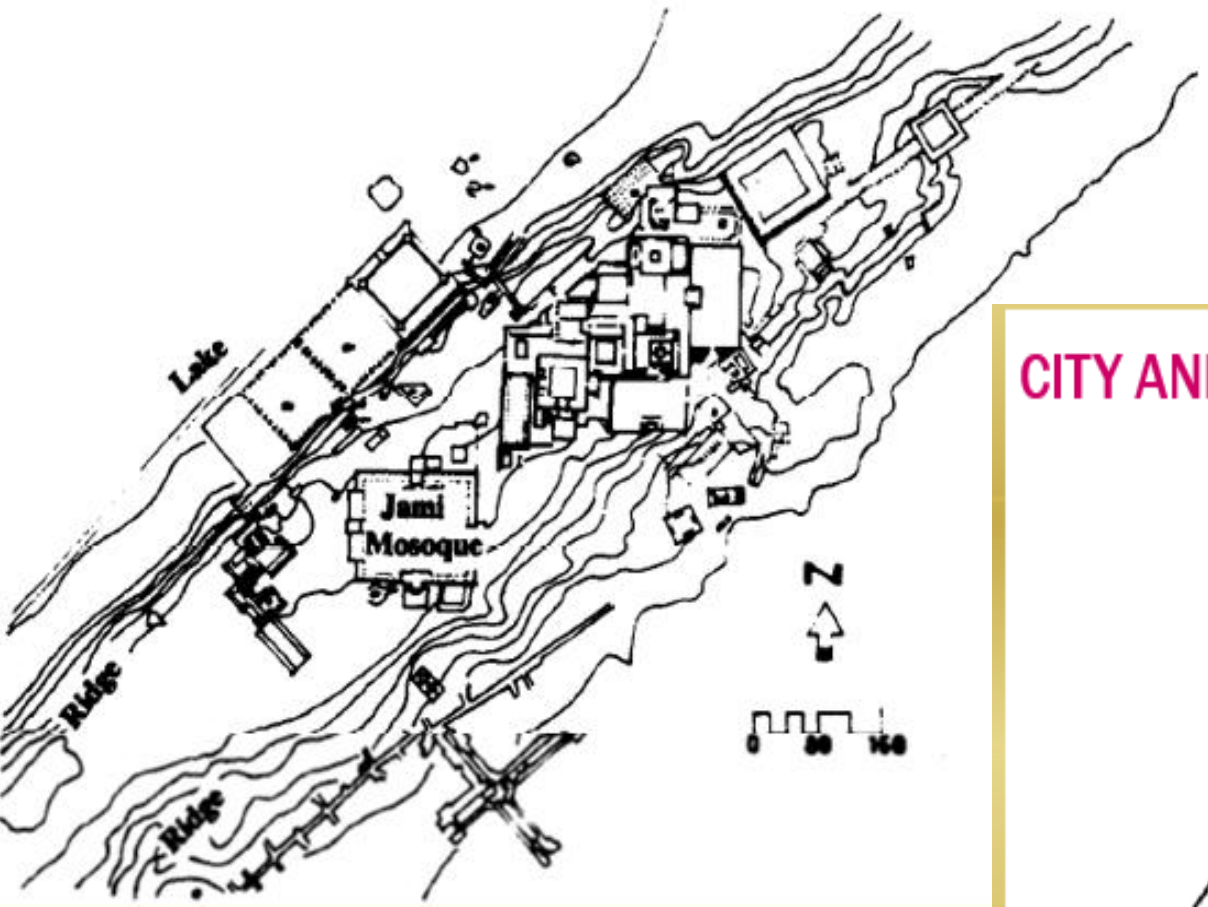
(1526 – 1707 A.D.)

PREPARED BY: PROF.D.V.PATEL

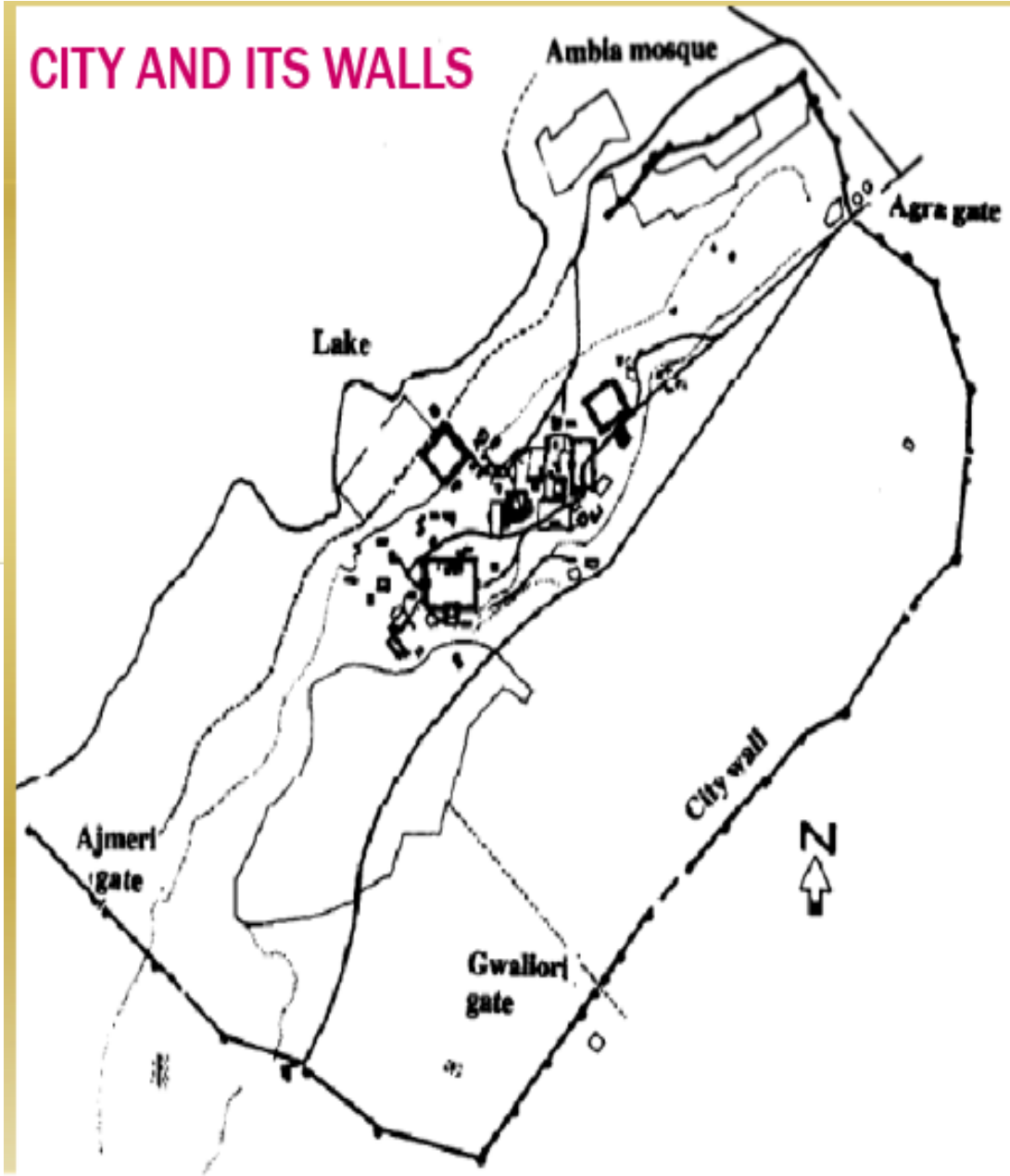
FATTEPUR SIKRI

- Founded by Akbar, this city has a long story, which says that Akbar built this city in the honour of the saint, Shaikh Salim Chishti. It is believed that, Akbar didn't had heir till the age of 26.
- Regarding this, he visited the saint, whose blessing gave Akbar 3 sons.
- Henceforth, the city of Red Sandstone buildings was established - the Fatehpur Sikri.
- Unlike Agra that wasd a thriving centre of trade, Sikri was a little village which had first come to Mughal notice when Babur, triumphant after defeating Rana Sanga at Khanuwa in 1527, according to a popular belief, named the village Shukri, meaning thanksgiving.
- Fatehpur Sikri is said to be the look-alike of the mosque in Mecca and has designs, taken from the Persian & Hindu architecture.
- It was planned as the cultural, commercial and administrative centre of mughal empire.Fatehpur Sikri is built in red sandstone, and is a beautiful blend of Hindu and Islamic architectural elements.
- Fatehpur Sikri was abandoned 14 years after its creation.
- Today it is a ghost city, its architecture is in a perfect state of preservation, and wandering through the palaces it is easy to imagine that this was once a royal residence and a dynamic cultural centre.

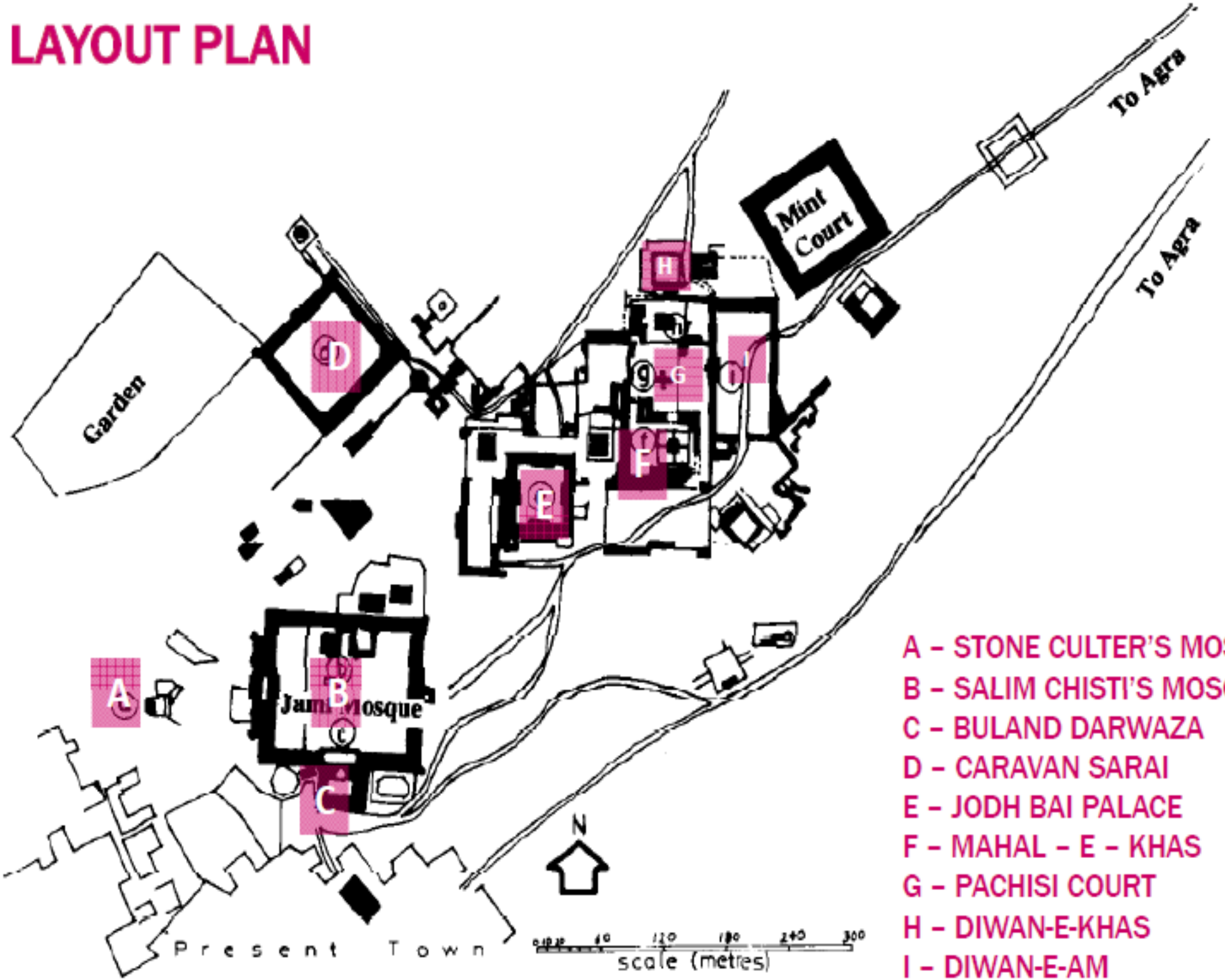
PREPARED BY: PROF.D.V.PATEL



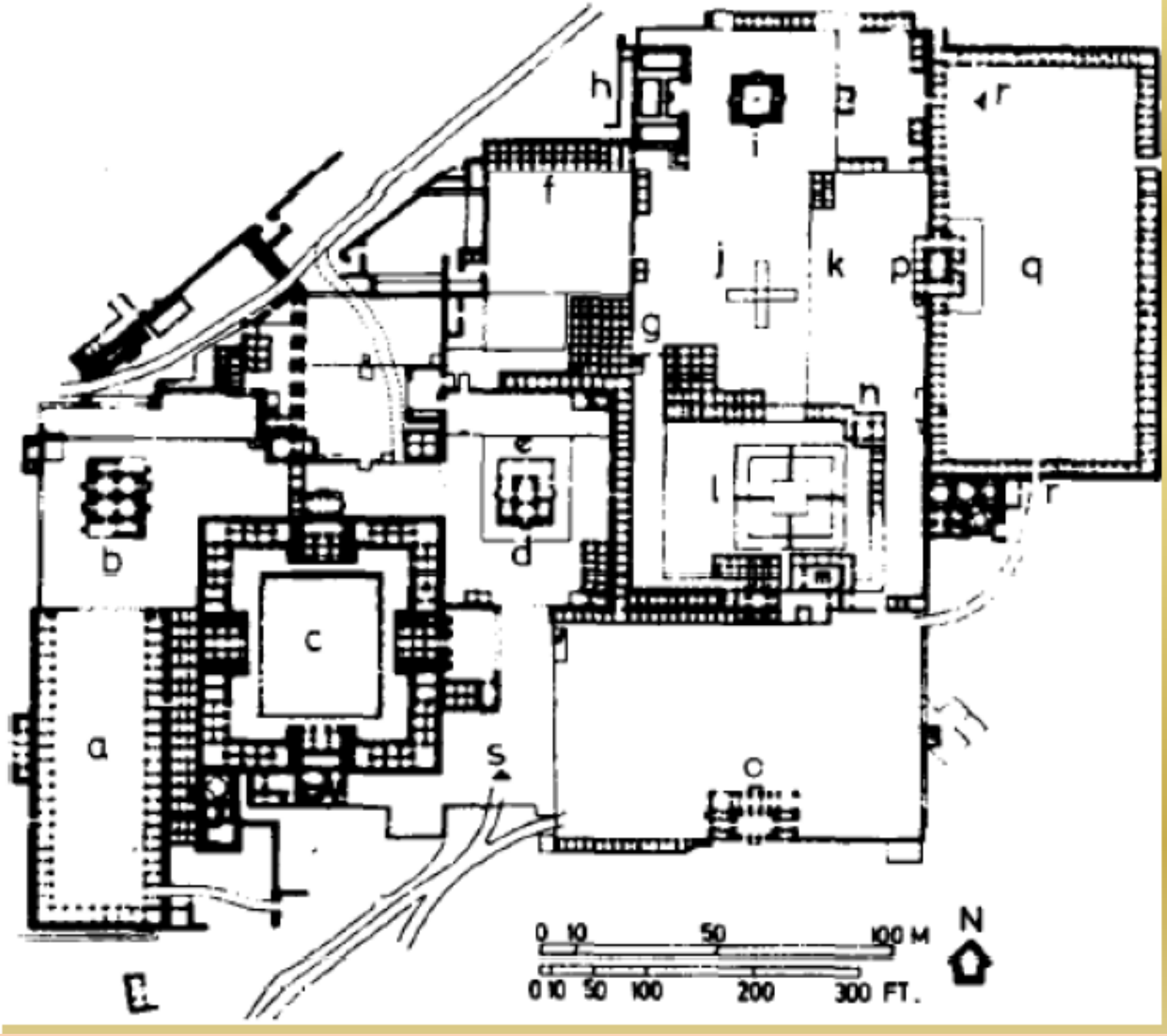
CITY AND ITS WALLS



LAYOUT PLAN



A – STONE CULTER'S MOSQUE
B – SALIM CHISTI'S MOSQUE
C – BULAND DARWAZA
D – CARAVAN SARAI
E – JODH BAI PALACE
F – MAHAL – E – KHAS
G – PACHISI COURT
H – DIWAN-E-KHAS
I – DIWAN-E-AM



A - STABLES FOR CAMELS & HORSES
B - RAJA BIRBAL'S HOUSE
C - JODH BAI'S HOUSE
D - MARYAM'S GARDEN
E - MARYAM'S HOUSE
F - HOSPITAL AND GARDEN
G - PANCH MAHAL
H - EMPEROR'S STUDY
I - DIWAN-E-KHAS
J - PACHISI COURT
K - GARDEN
L - EMPEROR'S PRIVATE APTS
M - EMPEROR'S SLEEPING QTZ
N - HOUSE OF THE TURKISH SULTANA
O - ADMINISTRATION AND ARCHIVES
P - DIWAN-E-AM PAVILION
Q - DIWAN-E-AM
R - ENTRANCES FOR PUBLIC
S - RESERVED ENTRANCE FOR IMPERIAL FAMILY

IMPERIAL COMPLEX: The imperial palace complex, consisting of the treasury, the offices, Daulat Khana, the Haram sara or ladies palace, now appears irregular and disjointed; but in Akbar's time is was part of a well connected and highly planned complex.

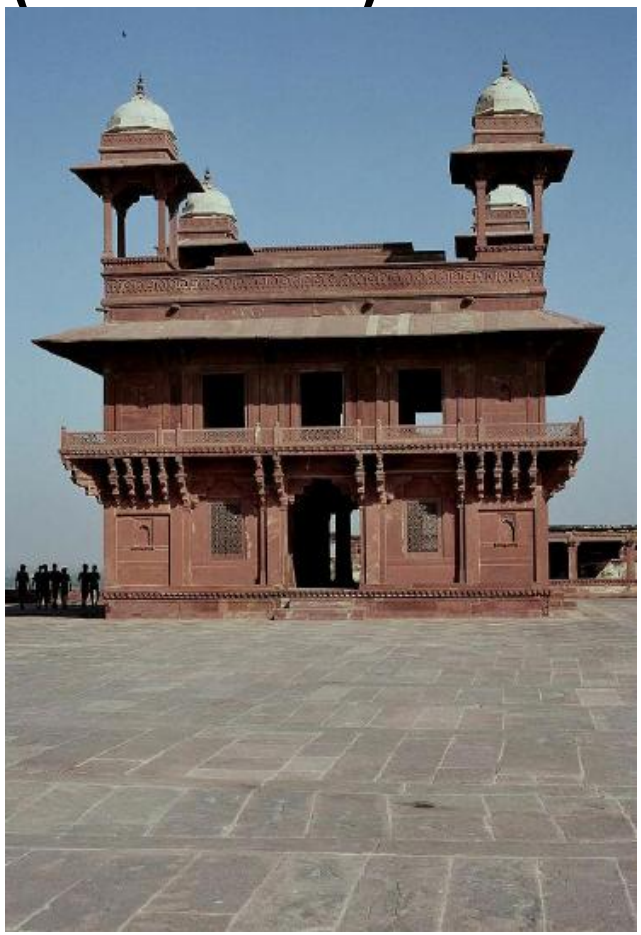
Diwan-i-Am (Hall of Public Audience) :



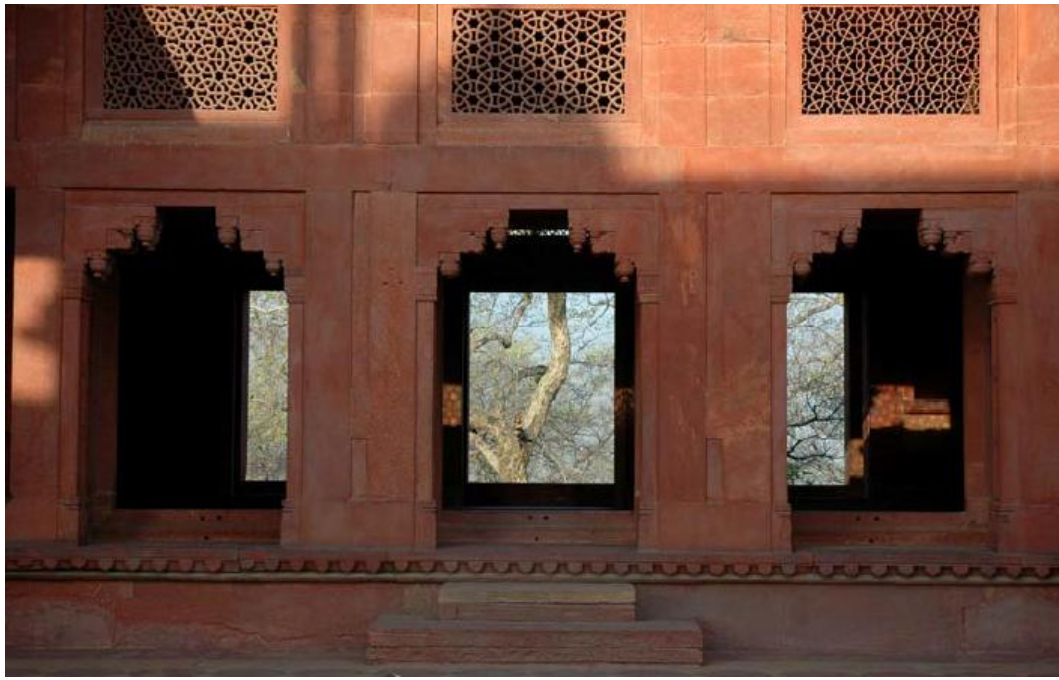


Diwan-i-Khass (Jewel House)

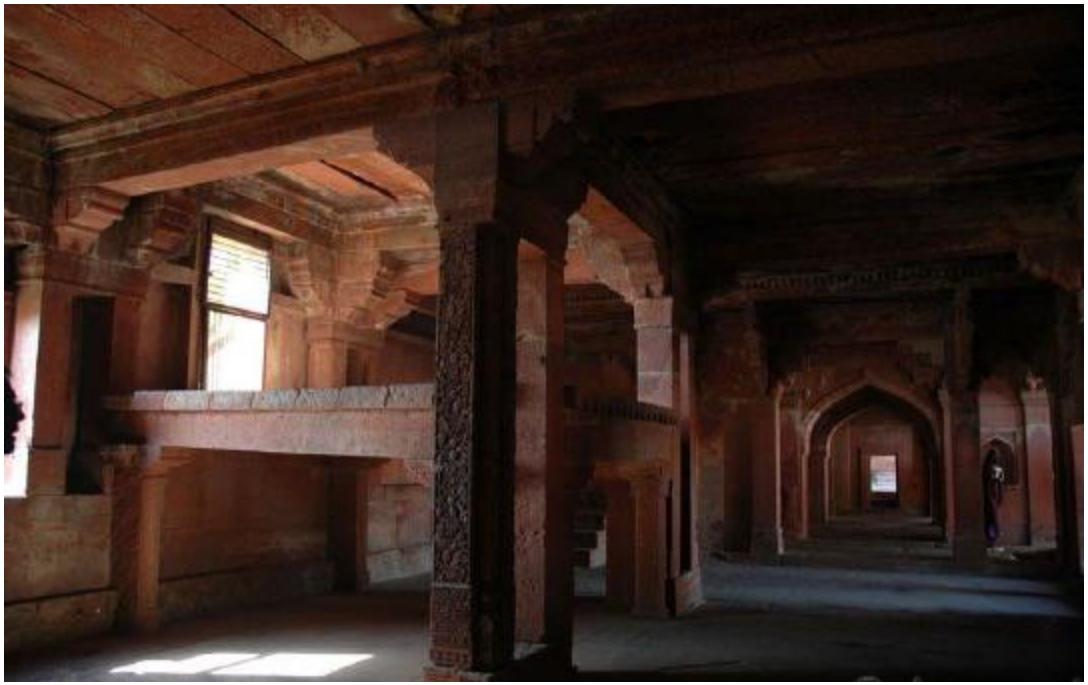
Diwan-i-Khas (Hall of Private Audience) : This is where Akbar met his ministers, scholars and generals to discuss religious and courtly matters.



Ankh Michauli



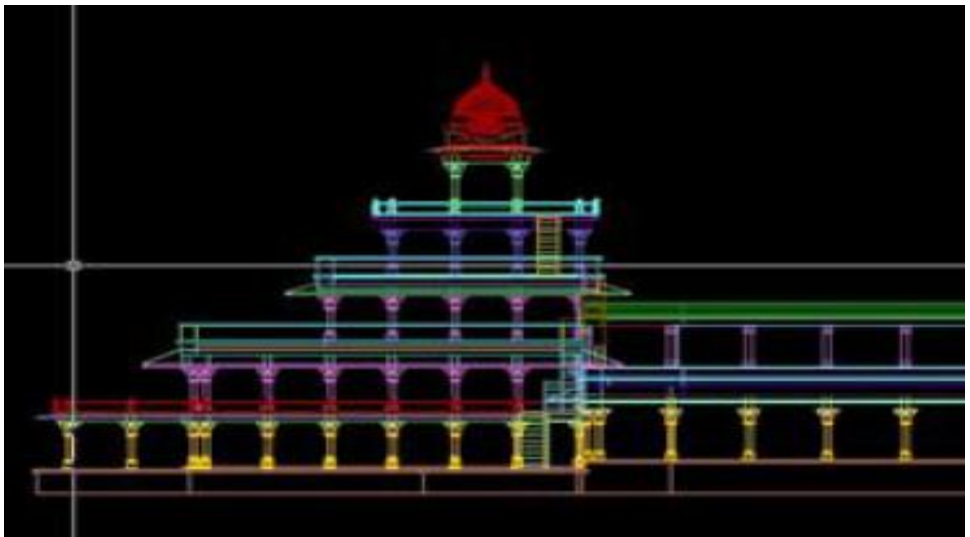
• **Diwan Khana-i-Khass :** Akbar is believed to have sat, discussing various issues with close advisors.



Pachisi Court : This is located at the very center of the main palace complex. On the open floor are the markings of a board game called Pachisi , hence the name.

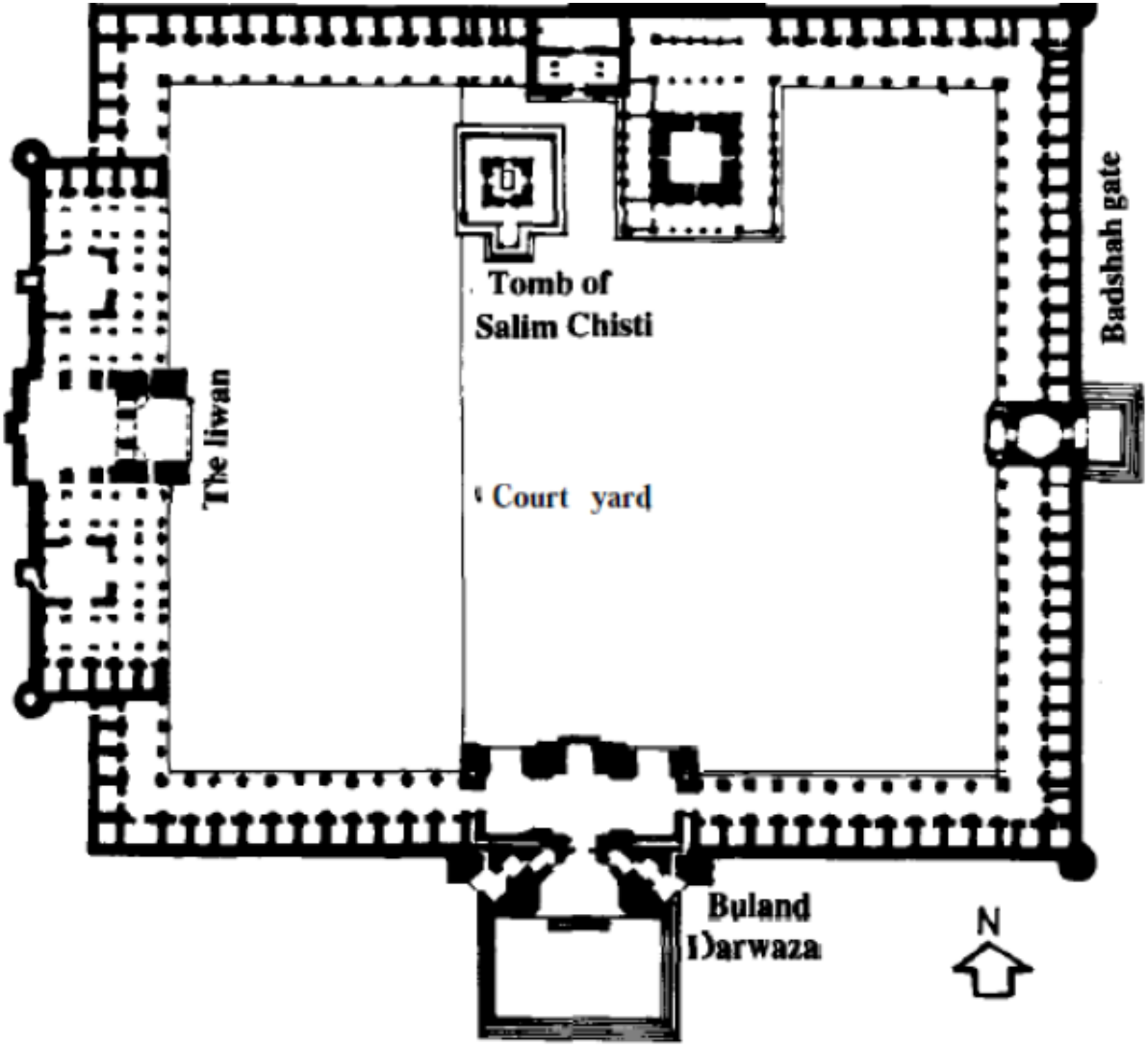


Panch Mahal: Panch Mahal is an extraordinary structure,entirely columnar, consisting of four storeys of decreasing size with a kiosk, disposed asymmetrically upon a ground floor that contains 84 columns.





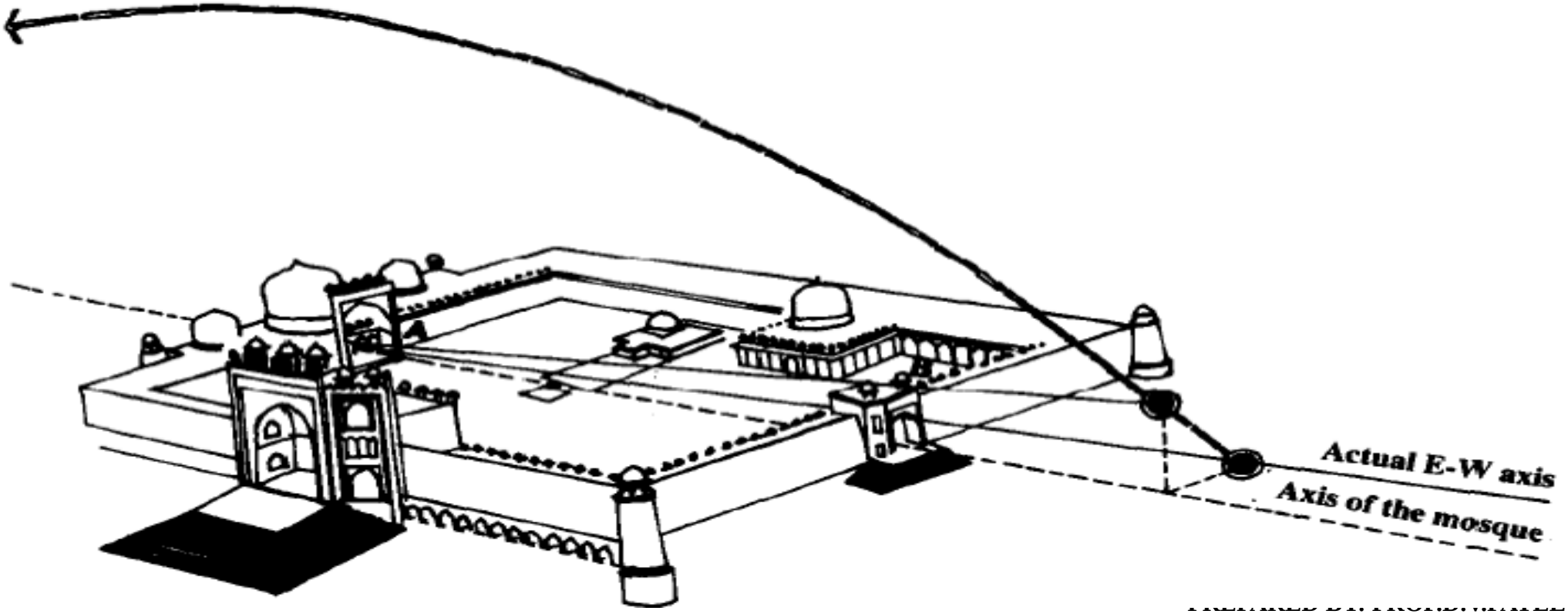
JAMI MOSQUE Q – AN EXPRESSION OF ARCHITECTURAL GRACE AND PERFECTION



Jami Masjid or the Friday Mosque is one of the most beautiful and largest mosques of the world. This elegant building is the most sacred building and the principal mosque of the town and thus situated on the highest point of the ridge



ACTUAL EAST WEST ORIENTATION IS A FEW DEGREES NORTH TO THE AXIS OF THE MOSQUE.AS A RESULT OF THIS AT SUNRISE AND DURING SPECIFIC PERIOD OF THE YEAR, THE SUN MOVES INTO A PARTICULAR POSITION OVER THE AXIS OF THE MOSQUE. IT REACHES THE REQUIRED ALTITUDE AND STRIKES THE CENTRAL MIHRAB



PRE-INDEPENDENCE PERIOD
OR
BRITISH PERIOD
(TILL 1947)

PREPARED BY: PROF.D.V.PATEL

- When the Britishers first settled in India they found most of the condition s of the towns to be unhealthy.
- So they started independent colonies on the outskirts of the existing towns.
- These extensions were known as 'Cantonments and Barracks' for the military occupied area and 'Civil Lines' for the residence of civilians and well-to-do people.
- Next they took to street planning.
- They adopted straight roads regardless of the cost or damages to social structure of the town.

•No large town planning scheme were taken till the end of 19th century.

• In the first decade of 20th century they took up the outstanding work of building New-Delhi.

• The plan was prepared on modern town planning principles by eminent town planner Edwin Lutyens assisted by Baker.

• The capital group of buildings like Government House, Council Hall, Secretariat has been designed with their monumental architecture, industrial buildings are separated from residential sector, and arranged around the commercial and civic buildings.

• It was more a planning and designing of Administrative centre than a new town as such.

Pre-Independence Period or British Period







Post Independence Period (After 1947)

PREPARED BY: PROF.D.V.PATEL

Many industrial towns were planned after post-independence period.

- **A few of these can be mentioned below.**

Steel Towns:

- Durgapur – West Bengal
- Bhilai – Madhya Pradesh
- Rourkela **-** Orissa

Industrial Towns:

- Jamshedpur – Bihar
- Bhadravati – Karnataka
- Chittaranjan – West Bengal

Capitals:

- Gandhinagar – Gujarat
- Chandigarh – Punjab (Now a union territory)

Background

1947

India's Partition

JAWAHARLAL NEHRU

"Let this be a new town, symbolic of freedom of India unfettered by the traditions of the past. An expression of the nations faith in the future"

New Capital City and Administration Centre accommodating half a million people and expandable to 1 million.

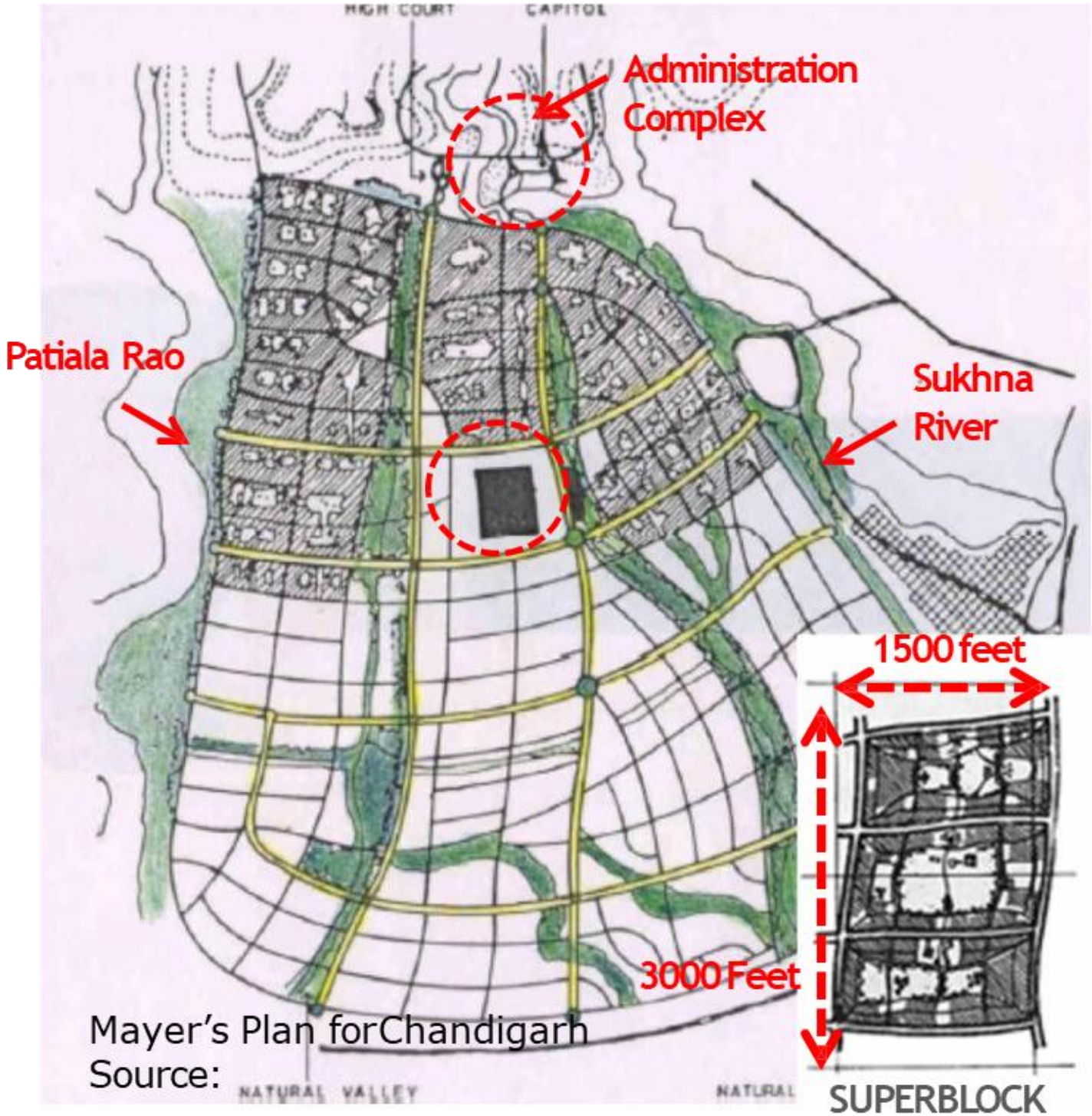
Mayer's Plan

Albert Mayer and Mathew Nowicky

Neighbourhood Unit Concept

- Located between two boundary rivers
- fan shaped plan
- Govt. Centre – Northeast Side of the City
- Super Block - Accommodating some 1500 families.
- 3 Superblocks to make a district
- Industrial area planned in southern corner
- Administration area in north-east
- 2 Large parkland zones across city
- Future expansion in the southwest across the plan

Criticism – Lacks Monumental Symbolism



Mayer's Plan forChandigarh
Source:

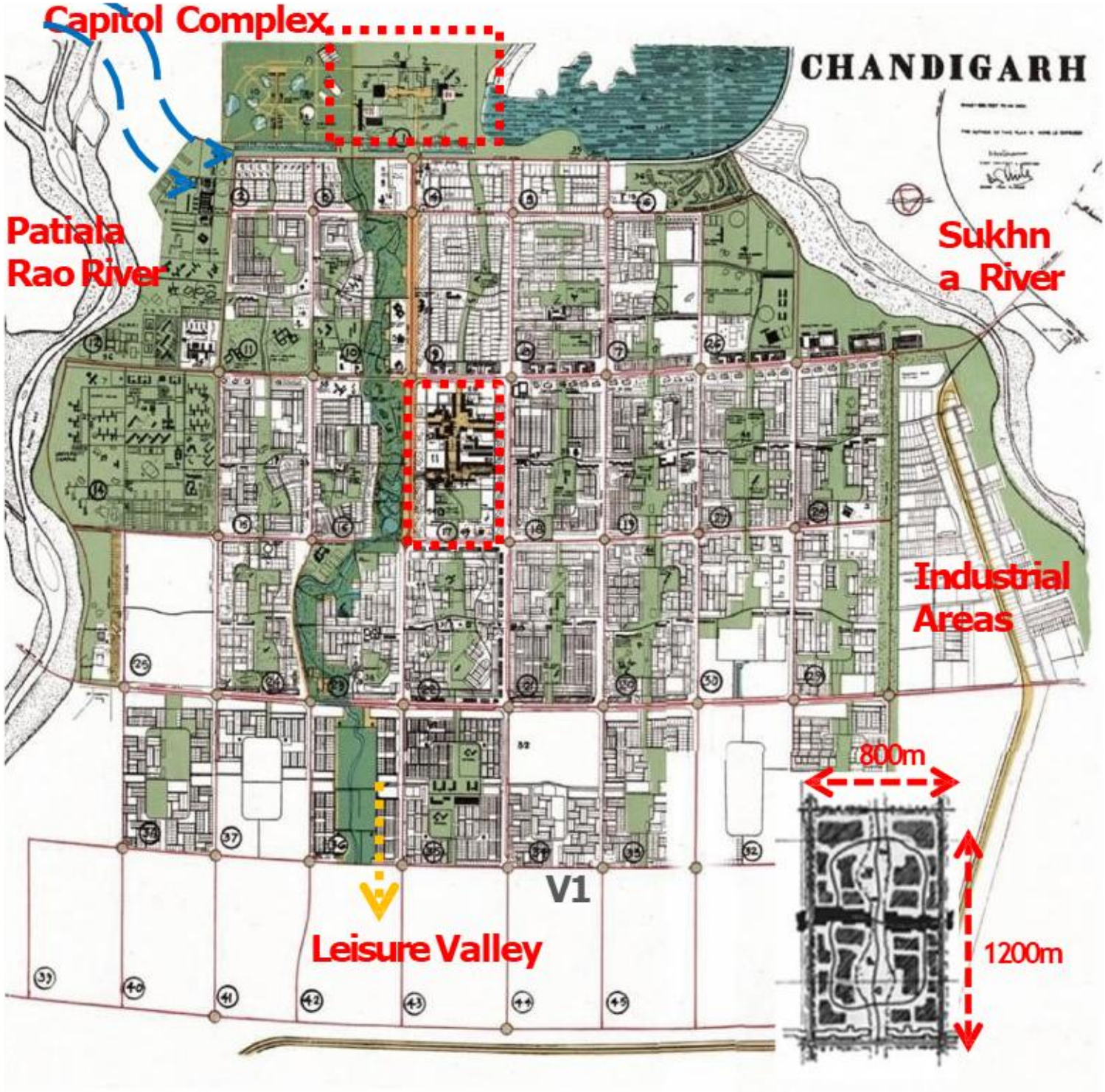
Turning Point

Mathew Nowicky's Death

Le Corbusier's Intervention

Formulation of new plan

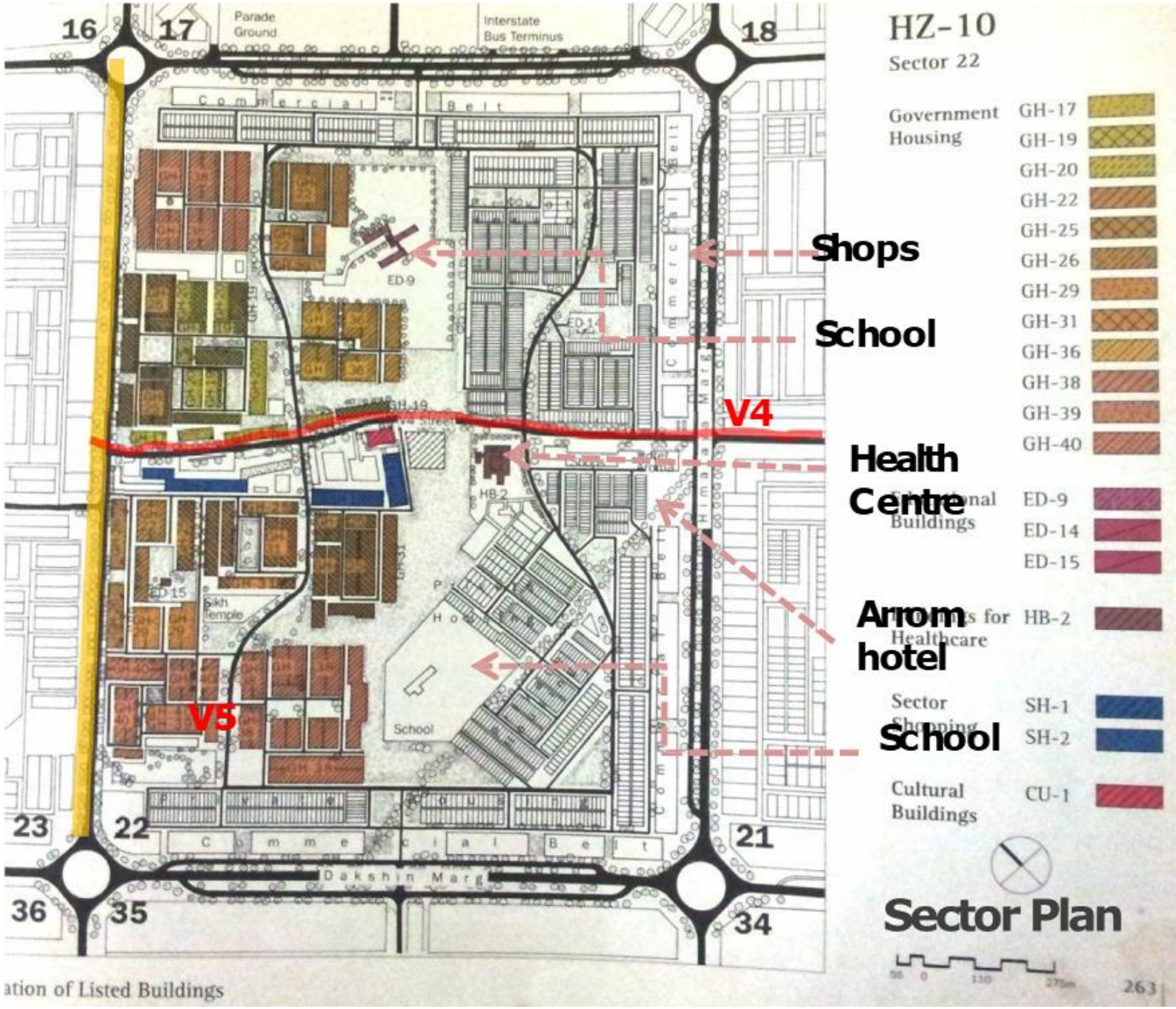
- Kept orientation of grids as it was
- One dominant off centred doglegged axis
- The grid of superblock increased in size to 1200 x 800mts (3900 x 2700 feet)
- Accommodating up to 25000 inhabitants
- Within superblock centralized market was replaced by continuous bazar street across the block.
- Capital Complex and high court combined in one govt. complex
- CBD Area Remained the same
- Shifted Capital complex to higher ground
- Two linear parks were replaced by total of 6 later by 8.
- Each Sector with its own green space around which houses are designed
- V7 Road Concept



PREPARED BY: PROF.D.V.PATEL

Sectors

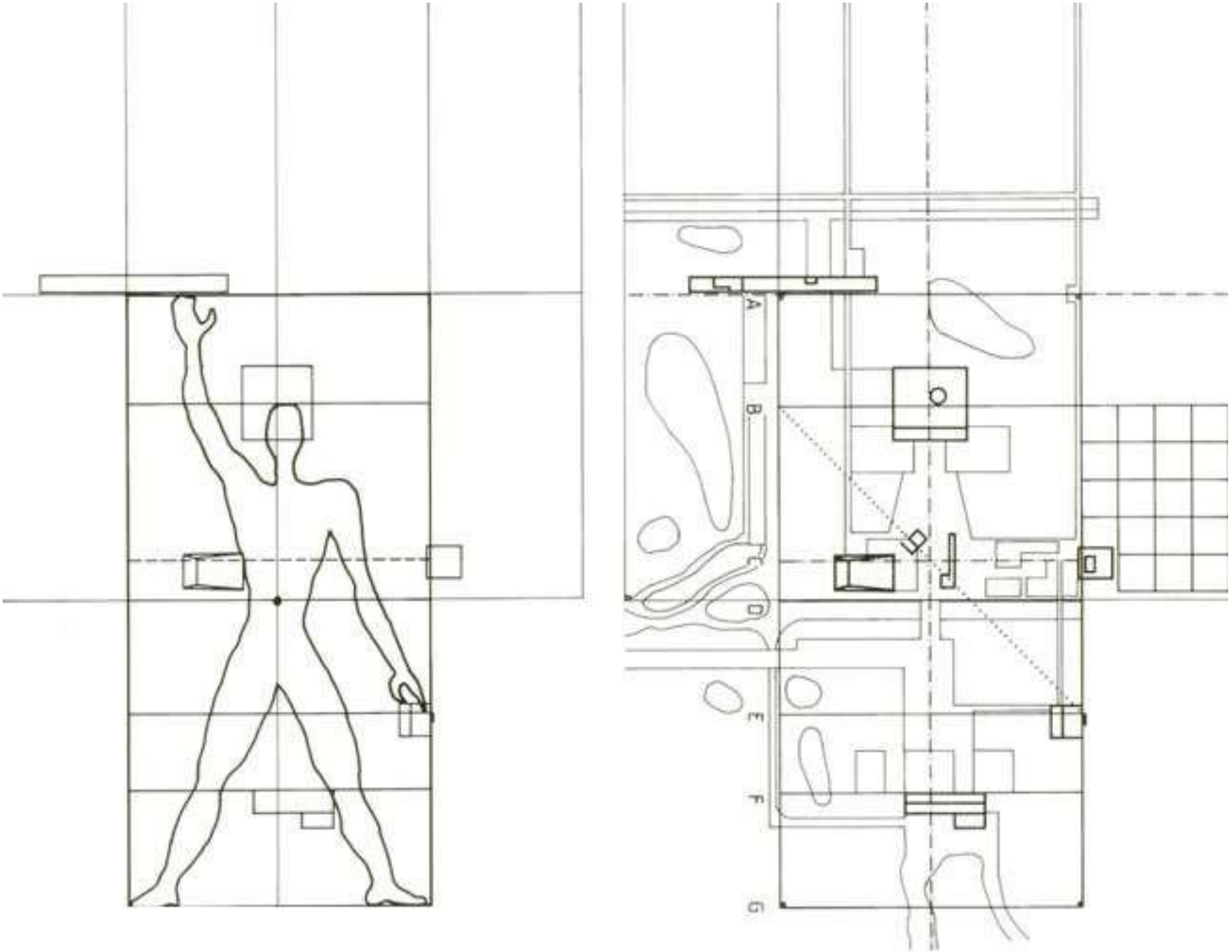
- Total 30 sectors - 24 residential
- 70% building private
- Autonomous unit including housing, school, shops, recreational spaces.
- 3000-25000 accommodating capacity
- Surrounded by high speed roads
- Bus stops every 400mts
- No pedestrian needs to walk more than 10 minutes



IDEA OF LE CORBUSIER

The city of Chandigarh is planned to human scale.

- Head–capitol (place of power)
- Heart –the city centre
- Stomach–the commercial area
- Arms–university and Industrial zone
- Lungs–leisure valley ,open spaces
- Arteries–network of roads

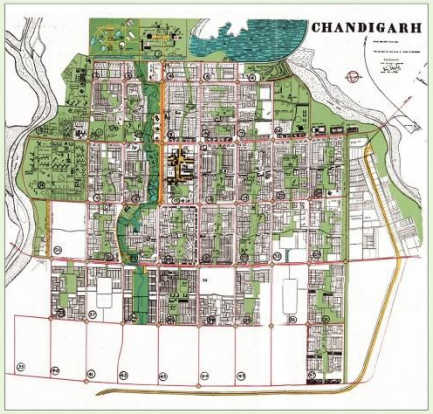

A
B
C
D
E
F
G

INTRODUCTION:-

CHANDIGARH was the first planned city **Designed By le Corbusier** after independence from British rule in 1947. It is the capital city of the states of Punjab and Haryana. The city is located at the picturesque junction of foothills of the Himalayas Mountain range and the Ganges plains. It houses a population of 1,054,600 inhabitants (2001) and is one of the richest cities of the nation.

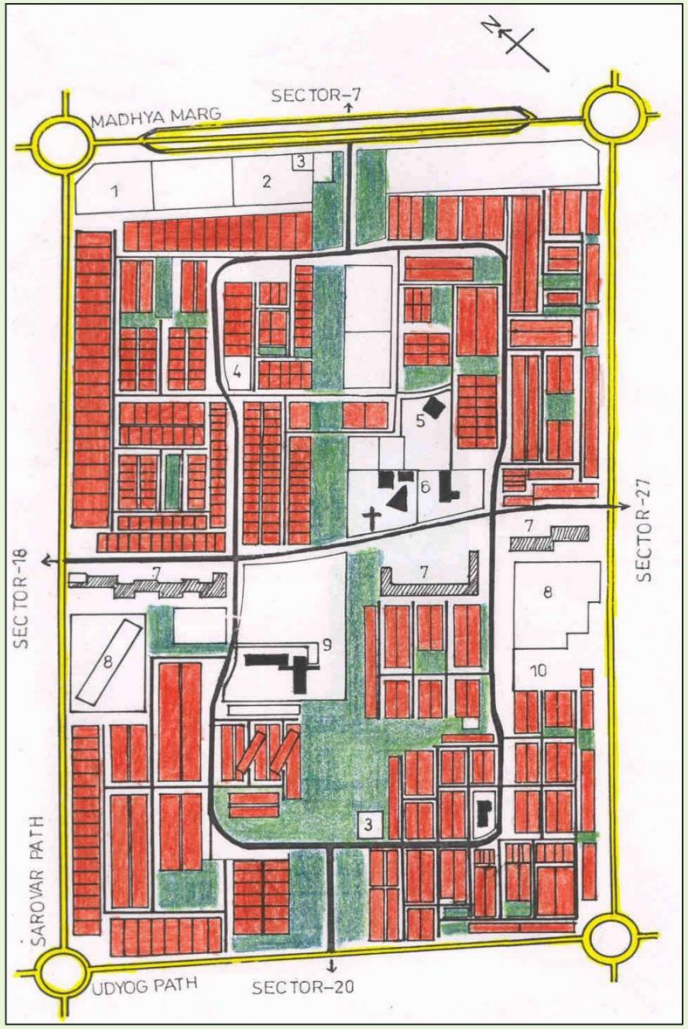
BASIC PLANNING CONCEPTS

Le Corbusier conceived the master plan of Chandigarh as analogous to human body, with a clearly defined
Head (the Capitol Complex, Sector 1),
Heart (the City Centre Sector-17),
Lungs (the leisure valley, innumerable open spaces and sector greens),
Intellect (the cultural and educational institutions),
Circulatory system (the network of roads,
the 7Vs) and Viscera (the Industrial Area)



Layout of a typical residential sector

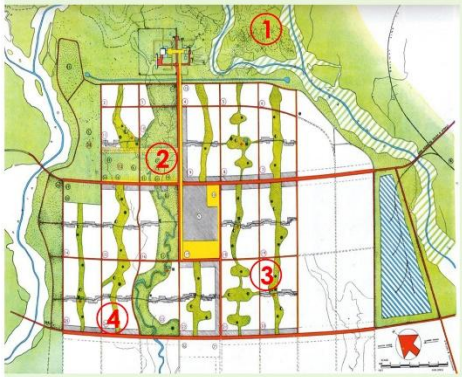
The primary module of city's design is a Sector, a neighborhood unit of size 800 m x 1200 m.



The basic geographical and demographic profile of Chandigarh is as under:

Area		114 sq kms
Longitude		760 47' 14E
Latitude		300 44' 14N
Altitude		304-365 m
Annual Rainfall (av)		1110.7 mm
Total Population (2001 census)		10,54,600
(Rural population-	92120	(10.2%)
(Urban population-	808515	(89.8%)
Density of population/sq. km.		7,900
Birth Rate (per 1000)		21.45 (2005)
Death Rate (per 1000)		10.22 (2005)
Infant Mortality Rate (per 1000)		44.13
Sex Ratio (females per 1000 males)		777
Decennial Population Growth		40.33%
Literacy Rate		81.9%

HIERARCHY of GREEN AREAS



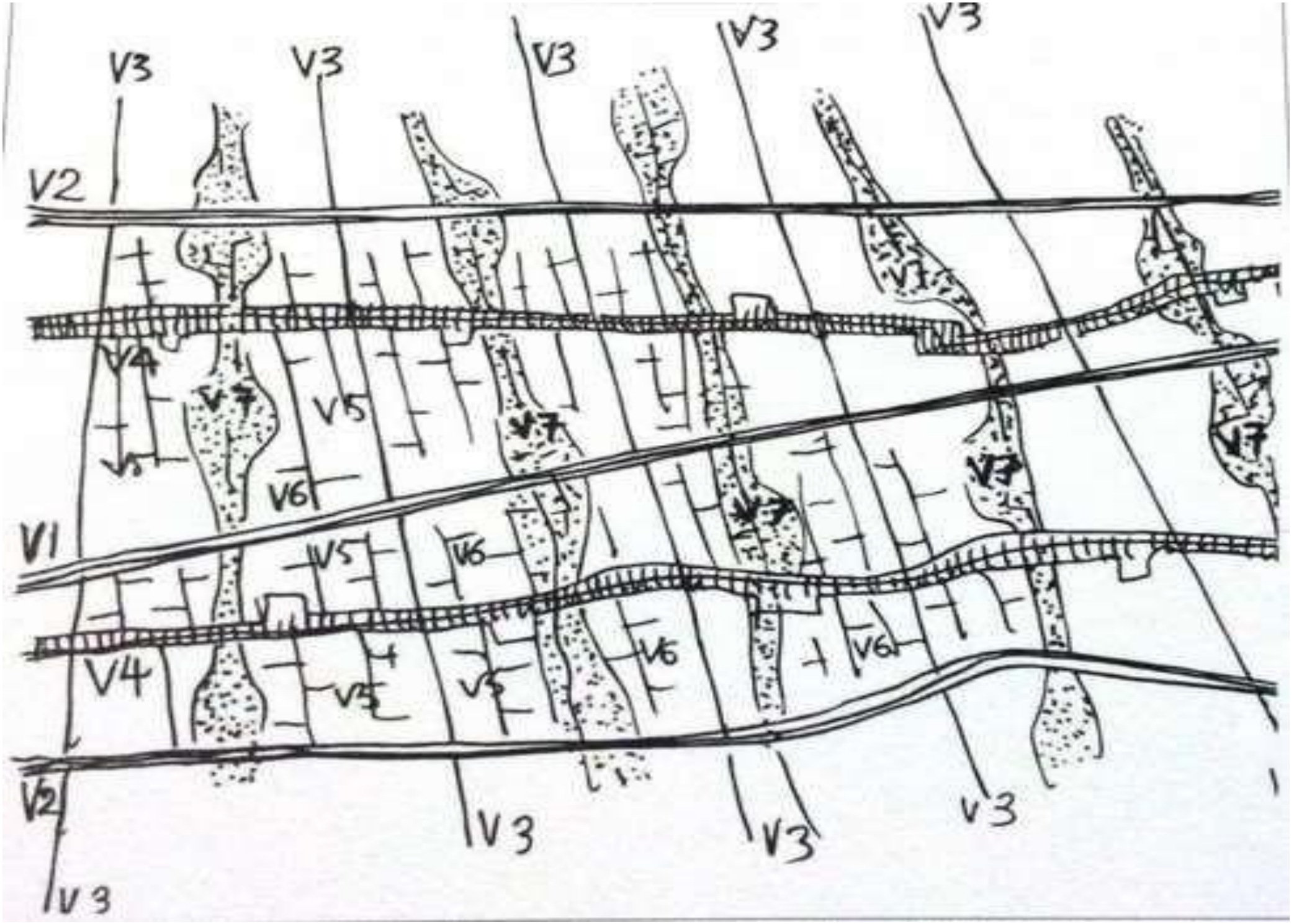
1 City Level Public Green Space with Artificial Water Body
2 Free- Flowing Green Space, connecting the entire site
3 Semi-Private Green Areas for neighborhood pockets
4 Private Green Areas for Residential Units

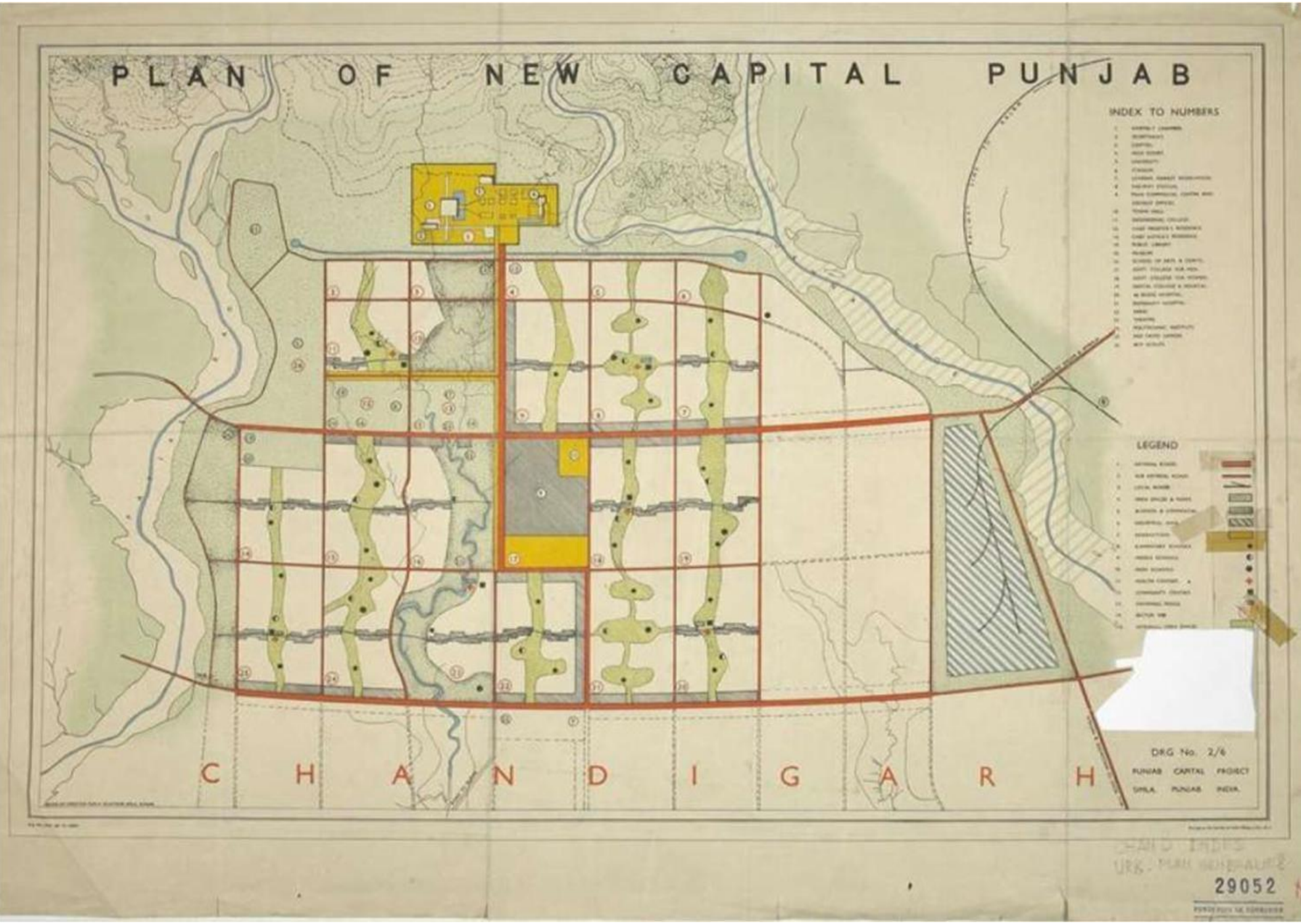
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ROAD SYSTEM

- The roads of the city were classified into seven categories, known as the system of 7 Vs.

- V-1 Fast roads connecting Chandigarh to other towns;
- V-2 Arterial roads;
- V-3 Fast vehicular roads;
- V-4 Free Flowing shopping streets;
- V-5 Sector circulation roads;
- V-6 Access roads to houses;
- V-7 Footpaths and cycle tracks

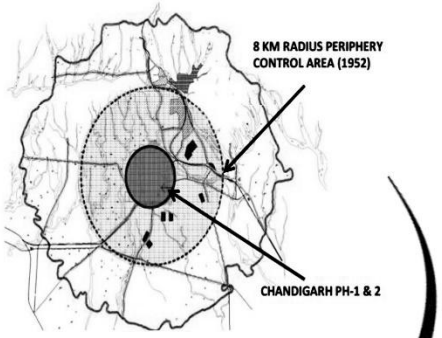

V3
V3
V3
V3
V3
V2
V4
V5
V6
V1
V5
V6
V4
V5
V3
V6
V6
V2
V3
V3
V3
V3

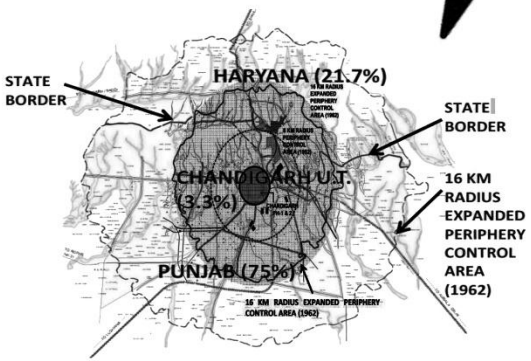


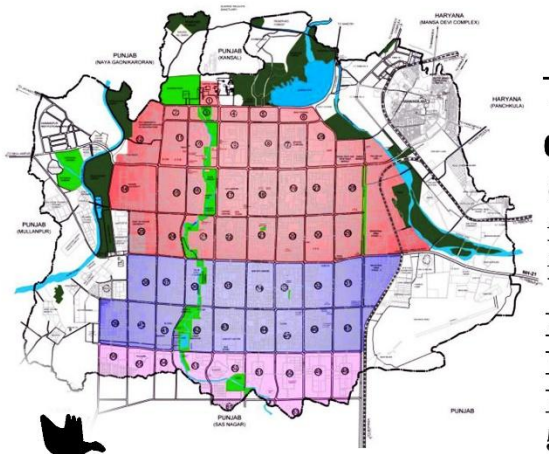
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Population Growth & Impact Analysis:-

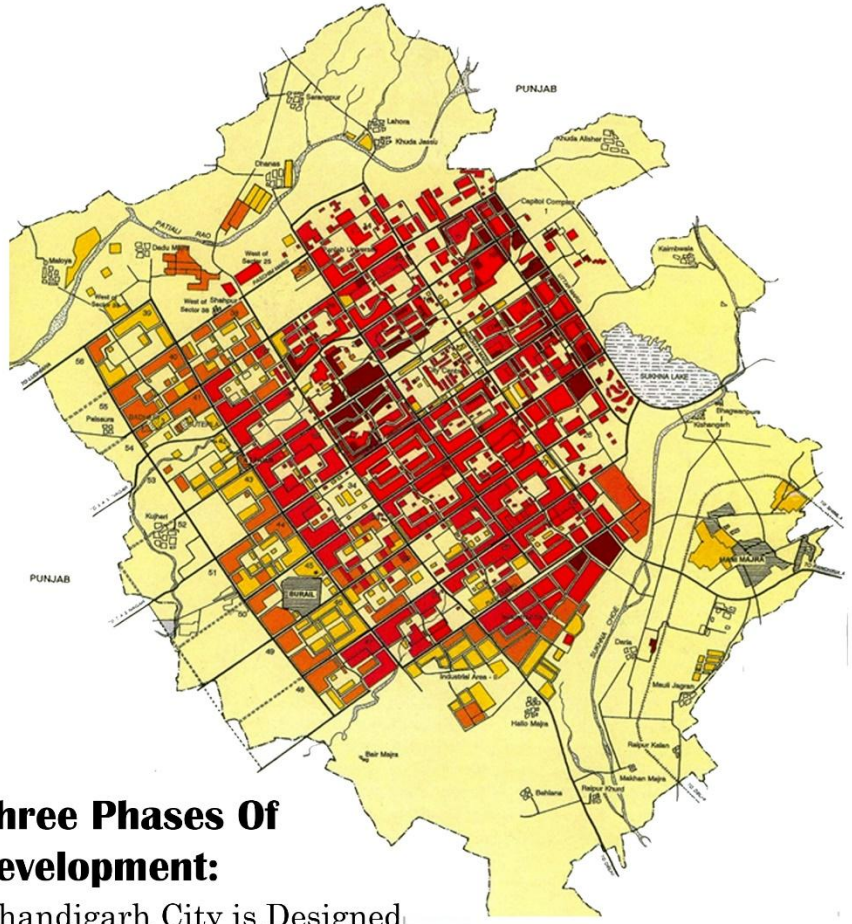
Expansion of City Pheriphery beyond 16km controlled area.







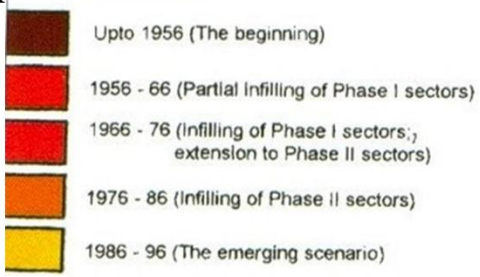
THE GROWTH OF CHANDIGARH 1966-96, STARTING WITH SECTORS 1 & 2 IN PHASE 1, & CURRENTLY CONSISTING OF 86 SECTORS



Three Phases Of development:

Chandigarh City is Designed in phases to tackle increase in population growth.

Phase1- Sector 1-30
Phase 2- Sector 31-47
Phase 3- Sector 48-56,61,63



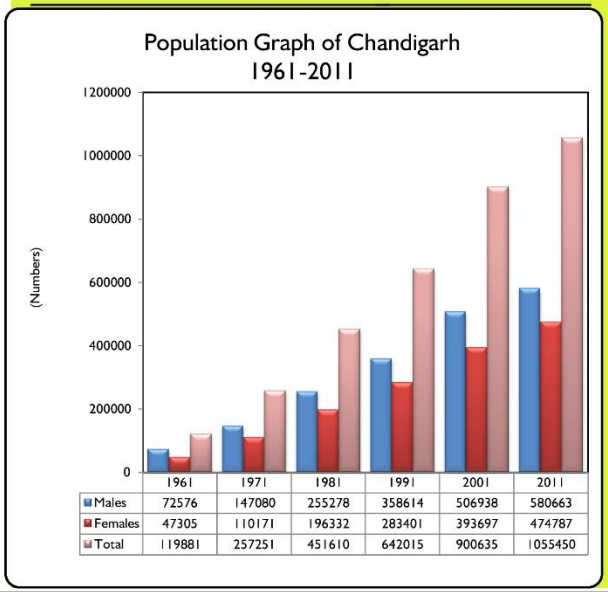
Factors For Rapid growth:

- Setting Of IT Park In the Northern Eastern sector of Maanimajra.
-Chandigarh serving as Hub of polital & bureaucratic activities of the neighbouring states(Punjab, Availabilty of Chandigarh & Himachal).
-High Profile education & Health facilities.
- High Development rate.

AREA AND POPULATION

Population of Chandigarh from 1961 to 2011 (census figures).
(Chandigarh was established in 1951).

Year	Total	Decennial Rate of Growth
1	2	5
1961	119881	-
1971	257251	+114.59
1981	451610	+75.55
1991	642015	+42.16
2001	900635	+40.28
2011	1055450	+17.19



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Assembly Building



Tower of Shadows



High court Building



Secretariat Building



Pedestrian Plaza



CAPITOL COMPLEX CHANDIGARH



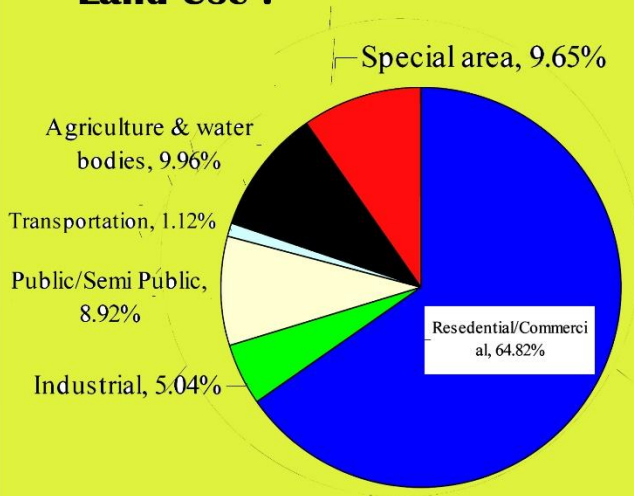
TIANANMEN SQUARE BEIJING

PREPARED BY: PROF.D.V.PATEL

DETAIL OF EXISTING LAND USE:-



Land Use :



DETAIL OF LAND USE

RESIDENTIAL
URBAN VILLAGES
VILLAGES IN PERIPHERY

COMMERCIAL
WHOLESALE

INDUSTRIAL
I.T. PARK

PUBLIC/ SEMI PUBLIC
INSTITUTIONAL
EDUCATIONAL
HEALTH FACILITIES
SPORTS FACILITIES
CREMATION GROUND
CULTURAL FACILITIES
PUBLIC UTILITES

GREEN / OPEN SPACES
ORGANISED OPEN SPA
FOREST
AGRICULTURE

TRAFFIC AND TRANSPORTATION
TRANSPORTATION NODES
ROADS
RAILWAY LINE

PREPARED BY: PROF.D.V.PATEL


Swastika Symbol

Open Hand



Source - Turner, T. (Director). (2015). *Chandigarh 1 (of 4): Le Corbusier's urban design and planning* [Motion Picture].

Trench of Consideration



Source - Turner, T. (Director). (2015). *Chandigarh 1 (of 4): Le Corbusier's urban design and planning* [Motion Picture].

